The research program of the Center for Economic Studies produces a wide range of theoretical and empirical economic analyses which serve to improve the statistical programs of the U.S. Bureau of the Census. Many of these analyses take the form of research papers. The purpose of the Discussion Papers is to circulate intermediate and final results of this research among interested readers within and outside the Census Bureau. The opinions and conclusions expressed in the papers are those of the authors and do not necessarily represent those of the U.S. Bureau of the Census. All papers are screened to ensure that they do not disclose confidential information. Persons who wish to obtain a copy of the paper, submit comments about the paper, or obtain general information about the series should contact Peter Zadrozny, Editor, Discussion Papers, Center for Economic Studies, Room 1587, FOB 3, U.S. Bureau of the Census, Washington, DC 20233 (301-763-4881).

ESTIMATING A MULTIVARIATE ARMA MODEL WITH MIXED-FREQUENCY DATA: AN APPLICATION TO FORECASTING U.S. GNP AT MONTHLY INTERVALS*

by

Peter A. Zadrozny**

CES 90-5 June 1990

ABSTRACT

This paper develops and applies a method for directly estimating a multivariate, autoregressive moving-average (ARMA) model with mixed-frequency, time-series data. Unlike standard, single-frequency methods, the method does not require the data to be transformed to a single frequency (by temporally aggregating higher-frequency data to lower frequencies or interpolating lower-frequency data to higher frequencies) or the model to be restricted by frequency. Subject to computational constraints, the method can handle any number of variables and frequencies. In addition, variables can be treated as temporally aggregated and observed with errors and delays. The key to the method is to view lower-frequency data as periodically missing and to use the missing-data variant of the Kalman filter.

In the application, a bivariate, ARMA model is estimated with monthly observations on total employment and quarterly observations on real GNP, in the U.S., for January 1958 to December 1978. The estimated model is, then, used to compute monthly forecasts of the variables for 1 to 12 months ahead, for January 1979 to December 1988. Compared with GNP forecasts, in particular, for similar periods produced by established econometric and time series models, present GNP forecasts are generally more accurate for 1 to 4 months ahead and about equally or slightly less accurate for 5 to 12 months ahead. The application, thus, shows that the present method is tractable and able to effectively exploit cross-frequency sample information, in ARMA estimation and forecasting, which standard methods cannot exploit at all.

*This work was mostly done while I was a Visiting Scholar at the Research Department of the Federal Reserve Bank of Atlanta, from October 1988 to November 1989. The views expressed here are my own and do not necessarily reflect any official positions of the Federal Reserve Bank of Atlanta or the Bureau of the Census. This work has greatly benefitted from conversations with William R. Bell, David F. Findley, and William T. Roberds. Shannon Mudd helped with some of the computations.

**CES, Bureau of the Census, Washington, DC

1. Introduction.

A general problem in the estimation of multivariate economic time-series models is that observations on different variables tend to be recorded and published at different frequencies. For example, in the U.S., data on gross national product (GNP), the unemployment rate, and the money stock (e.g., M2) are, respectively, available at quarterly, monthly, and weekly sampling intervals. Therefore, virtually any attempt to estimate a multivariate economic time-series model confronts the problem of efficiently using mixed-frequency data. Standard time-series estimation methods are generally not satisfactory in this respect because they are designed for data at a single frequency. When using a single-frequency method with mixed frequency data, one has three options: (i) to transform the data to a single frequency by temporally aggregating higher-frequency observations to the lowest frequency in the data; (ii) to transform the data to a single frequency by interpolating lower-frequency observations to the highest frequency in the data; or (iii) to causally order the variables by frequency, with higher-frequency variables being causally prior to lower-frequency variables, and suitably restricting the autoregressive part of the model by frequency, so that each causal block of the model can be estimated separately, at its own frequency (e.g., Fitzgerald and Miller 1989, Trehan 1989).

Neither of these three options is generally satisfactory. First, temporal aggregation destroys sample information. Second, commonly used interpolation methods generally do not fully exploit the available sample information. For example, in statistics, it is common to interpolate time series separately, using only their serial correlations

(e.g., Harvey and Pierse 1984). In economics, regresssion methods, which emphasize correlations across variables, are favored, but even the most advanced regression-interpolation method which exploits serial correlations in residuals (Litterman 1983) will generally be unable to exploit all significant cross-serial correlations. When the goal is to estimate a model, for forecasting or other purposes, interpolation is at best an intermediate nuisance and at worst a source of distortion in the data to be used in estimation. Third, the consequences of inappropriate causality and other restrictions, of course, depend on the application. For example, as we shall demonstrate, such restrictions can markedly degrade the forecasting abilities of a model.

This paper describes and illustrates a method for directly estimating a multivariate, autoregressive moving-average (ARMA) model with mixed-frequency, time-series data. Unlike standard, single-frequency methods, the method does not require the data to be transformed to a single frequency or the model to be restricted by frequency, in particular, causally ordered by frequency. Subject to computational constraints, the method can handle any number of variables and frequencies. In addition, variables can easily be treated as temporally aggregated and observed with errors and delays.

The approach of the method is to assume that the model operates at the highest frequency in the data. All variables are assumed to be generated, but not necessarily observed, at this highest frequency. Variables which are observed at lower frequencies are viewed as being periodically missing. For example, in monthly-quarterly data, the model is assumed to generate all variables at monthly intervals and each quarterly observation is assigned to a month in that quarter (the month

can vary from variable to variable and quarter to quarter), so that observations for the remaining months in the quarter are viewed as missing. Then, the Kalman filter (KF), modified to handle missing observations (Jones 1980, Ansley and Kohn 1983), is used to compute the Gaussian (or normal distribution) likelihood function in maximum likelihood estimation. The modification basically involves skipping computations in the standard KF which are associated with the missing observations. The method allows one to handle any set of frequencies, indeed, any pattern of missing data, subject only to the data having enough information to identify the parameters (estimate them with finite precision).

In the application, a bivariate ARMA model is estimated with monthly observations on total employment and quarterly observations on real GNP, in the U.S., for the period January 1958 to December 1978. The estimated model and the modified KF are, then, used to compute monthly forecasts of GNP and total employment, for 1 to 12 months ahead, for the period January 1979 to December 1988. Several low-order models were considered (seasonally adjusted data were used). The ARMA(1,1) model was judged to be the best model, in terms of maintaining a balance among the criteria of: (a) minimizing a bias-corrected Akaike information criterion (Hurvich and Tsai 1989), (b) providing the most accurate GNP forecasts (in terms of lowest overall standard errors of forecasts for 1 to 12 steps ahead), (c) eliminating serial correlations of innovations (residuals), and (d) having the most parsimonious parameterization. Accuracy of GNP forecasts, in particular, was emphasized, because the novelty is the ability of the method to produce monthly forecasts of quarterly data with a general, monthly, ARMA model.

The application shows that the method is computationally tractable. With sufficient experience, a similar exercise can be conducted in an afternoon on a personal computer. The application also shows that the method can exploit significant cross-frequency sample information (cross-serial correlations between variables observed at different frequencies) which standard single-frequency methods cannot exploit at all. First, whereas the data (in the form of annualized percentage changes) had significant own- and cross-serial correlations for lags up to about 6 months, innovations of the best model displayed almost no significant serial correlations. Second, the GNP forecasts of the model compare well with those for similar periods produced by established econometric and vector autoregressive models (McNees 1986, Roberds 1988). Generally, present forecasts were more accurate for 1 to 4 months ahead and about equally or slightly less accurate for 5 to 12 months ahead. To provide a more direct evaluation of the ability of the method to exploit cross-frequency information, forecasts are also developed with separate, univariate models and with a model restricted in the manner of option (iii) above. Compared with the best results, the latter forecasts are notably less accurate for 1 to 4 months ahead and are about equally accurate for 5 to 12 months ahead. The results are encouraging for the present method, because they show that it can produce superior short-term and competitive medium term forecasts with a small information set (observations on few variables).

The KF also makes it easy to account for temporal aggregation, precision of data (observation or measurement errors), and timeliness of data (production delays), complications which are difficult to handle with non-KF methods. Here, GNP is treated as temporally aggregated, as a

moving sum of three consecutive monthly values. This is the natural thing to do because GNP is a flow which accrues over a period of time. In fact, doing so markedly improved the fit and forecasting accuracy of the model. Covariances of observation errors were set by calibration and the (meager) available information about the relative precisions of the data. Introducing observation-error covariances in this way improved the results of estimation and forecasting.

Unlike in Conrad and Corrado (1979), Howrey (1984), or Scadding (1987), no attempt was made here to account for the different precisions of preliminary and revised data by modelling their observation errors. Nor, unlike in Bordignon and Trivellato (1989), was any attempt made to account for timeliness of data, as one should in a real-time exercise. Simply, the most up-to-date historical record, which mixes extensively revised earlier values with preliminary recent values, was used. In the aforementioned studies, the standard KF was applied to single-frequency data. It would be useful to extend the present application by similarly accounting for precision and timeliness of data.

We also note two pertinent sets of studies. First, the question of whether preliminary GNP data should be construed as forecasts or as error-corrupted observations of true GNP has recently been debated extensively, but inconclusively (e.g., Mankiw and Shapiro 1986, Mork 1987, Scadding 1988). This debate has been conducted outside of the framework of the KF. Second, Corrado and co-researchers (see, e.g., Corrado and Greene 1988 and references therein) have applied the theory of optimal linear combinations of forecasts to update quarterly forecasts of GNP, and other aggregates for the U.S. economy, with monthly forecasting information. They interpret the method of combining

forecasts in terms of a nonrecursive KF.

The paper proceeds as follows. Section 2 introduces the general ARMA model and a state-space representation of it which is convenient for estimation and forecasting. Section 3 discusses how to compute the Gaussian likelihood function with the modified KF. Although the material in sections 2 and 3 is familiar (e.g., Zadrozny 1988, 1989), it is included to make the paper self-contained. Section 4 discusses the application and section 5 concludes the main text with remarks. An appendix discusses restrictions on parameters under which the KF converges and is numerically stable. Convergence and numerical stability of the KF are generally necessary for the computations to be tractable and feasible, and, therefore, determine the practical limitations of the present method.

2. State-Space Representation of the ARMA Model.

The discussion in this section proceeds at a general level except in the treatment of frequencies of observation. Although, to avoid cumbersome notation, we only explicitly discuss the monthly-quarterly case of the application, the generalization to any number of frequencies of observation should be apparent.

We assume that the model operates at the highest available frequency in the data, in this case monthly. Let the $n \times 1$ vector $u(t)$ collect the n variables of the model. Because the data are assumed to have been adjusted for mean values and other possible fixed (regression) effects, we write the general, monthly, ARMA(p,q) model for $u(t)$ as

$$A(L)u(t) = B(L)e(t), \tag{2.1}$$

for $t = 1, \ldots, N$, where $A(L) = A_0 - \Sigma_{k=1}^{p} A_k L^k$, $B(L) = \Sigma_{k=0}^{q} B_k L^k$, L is the monthly lag operator (e.g., $Lu(t) = u(t-1)$), and e(t) is an $n \times 1$, unobserved, Gaussian (normally distributed), white-noise (serially uncorrelated), disturbance vector with zero mean and constant, covariance matrix. The distributional assumptions on e(t) are denoted by $e(t) \sim NIID[0, \Sigma_e]$, where $\Sigma_e = Ee(t)e(t)^T$ (superscript T denotes transposition). In general, especially in models motivated by an economic theory, the coefficient matrices, $A_0, \ldots, B_q$, will be restricted in terms of a smaller number of structural parameters, collected in the vector ϕ.

The coefficient matrices may assume any values in $\mathbb{R}^{nxn}$, subject to the following restrictions. First, we require that the model be complete (produce a unique realization of u(t) for given values of past u's and current and past e's). Therefore, we assume that A_0 is nonsingular. Second, we assume that any identities in the data have been removed, so that the model implies a nonsingular, probability distribution of the data. One of the conditions which usually helps to ensure this result, and which we assume, is that $B_0\Sigma_e B_0^T > 0$ (positive definite). The other conditions are discussed in section 3. Third, there is redundancy (an identification problem) among A_0, B_0, and Σ_e which needs to be eliminated with some normalization. We adopt the normalization $A_0 = I_n$, B_0 = lower triangular, and $\Sigma_e = I_n$, where I_n is the $n \times n$ identity matrix. This normalization is convenient because, given $\Sigma_e = I_n$, $B_0\Sigma_e B_0^T > 0$ when all elements on the principal diagonal of B_0 are nonzero.

A state-space representation of a linear, dynamic model has two major parts: (i) a state vector and its law of motion, which comprise

the relevant information for forecasting the process in question; and (ii) an observation (or measurement) equation which tells how observations of the process are made in terms of the state vector. We now develop for (2.1) a state vector and law of motion.

By analogy with a continuous-time analysis (Zadrozny 1988), we define stocks as variables which are observed at the fundamental, monthly, sampling interval and define flows as variables which are observed as sums (temporal aggregates) over several monthly intervals. Partition $u(t) = [u_1(t)^T, u_2(t)^T]^T$, where $u_1(t)$ is an $n_1 \times 1$ subvector of stocks and $u_2(t)$ is an $n_2 \times 1$ subvector of flows, so that $n_1 + n_2 = n$. Let $w(t)$ be the $m \times 1$ vector of potential observations on $u(t)$. By "potential" we mean that $w(t)$ is not yet adjusted for the fact that some or all of its elements may be unobserved (missing) in period t. Also, partition $w(t) = [w_1(t)^T, w_2(t)^T]^T$, where $w_1(t)$ is an $m_1 \times 1$ subvector of stocks and $w_2(t)$ is an $m_2 \times 1$ subvector of flows, so that $m_1 + m_2 = m$. Note that $m_1 \leq n_1$ and $m_2 \leq n_2$, hence, $m \leq n$, the strict inequalities holding when some stocks or flows are observed as cross-sectional aggregates.

Abstracting for the moment from observation errors, stocks are observed as $w_1(t) = u_1(t)$ and flows are observed as $w_2(t) = \sum_{k=0}^{\upsilon} C_k u_2(t-k)$, where the C_k's are $n_2 \times n_2$, diagonal, indicator matrices, with zeroes and ones on the principal diagonal, and υ is the maximum lag in any flow observation. By setting leading $C_{k,ii}$'s $= 0$, where $C_{k,ii}$ is the (i,i) element of C_k, each flow can be observed with its own delay, and, by setting trailing $C_{k,ii}$'s $= 0$, each flow can have its own degree of temporal aggregation. In this scheme, a variable which is observed directly (not temporally aggregated), but with delay, is treated as a

"flow."

Let $x(t)$ be the $s \times 1$, state vector of the form $x(t) = [x_1(t)^T, \ldots, x_r(t)^T]^T$, where $r = \max(p, q + 1, \nu)$ and $x_k(t)$, for $k = 1, \ldots, r$, is $n_* \times 1$, where $n_* = n_1 + 2n_2$, so that $s = n_* r$. Let the $n_* \times n_*$ and $n_* \times n$ matrices A_k^* and B_k^* be defined by

$$(2.2) \qquad A_k^* = \begin{bmatrix} A_k^{11} & A_k^{12} & 0 \\ A_k^{21} & A_k^{22} & 0 \\ 0 & C_k & 0 \end{bmatrix}, \quad B_k^* = \begin{bmatrix} B_k^1 \\ B_k^2 \\ 0 \end{bmatrix},$$

where A_k^{ij} is quadrant (i,j) of A_k and B_k^i is block-row i of B_k, conformable with the partition $u(t) = [u_1(t)^T, u_2(t)^T]^T$. Let the law of motion of $x(t)$ be

$$(2.3) \qquad x(t) = Fx(t-1) + Ge(t),$$

where $$F = \begin{bmatrix} A_1^* & I & 0 & \cdots & 0 \\ \vdots & 0 & \ddots & \ddots & 0 \\ \vdots & \vdots & \ddots & \ddots & I \\ A_r^* & 0 & \cdots & \cdots & 0 \end{bmatrix}, \quad G = \begin{bmatrix} B_0^* \\ \vdots \\ B_{r-1}^* \end{bmatrix},$$

$A_k^* = 0$ for $k > \max(p, \nu)$, $B_k^* = 0$ for $k > q$, and the zero and identity matrices in F are all $n_* \times n_*$. F is called the state transition matrix and is in companion form when it has the above sparse pattern.

To see that $x(t)$ and (2.3) constitute a valid state vector and law of motion for (2.1), multiply out (2.3), to obtain the representative equation

$$(2.4) \qquad x_k(t) = A_k^* x_1(t-1) + x_{k+1}(t-1) + B_{k-1}^* e(t),$$

for $k = 1, \ldots, r$, where $x_{r+1}(t) = 0$, for all t. Use (2.4), for $k = r$, to eliminate $x_r(t-1)$ in (2.4), for $k = r - 1$. Then, use the result to eliminate $x_{r-1}(t-1)$ in (2.4), for $k = r - 2$. Continue in this fashion until (2.4), for $k = 1$, is

$$(2.5) \qquad x_1(t) = A_1^* x_1(t-1) + \ldots + A_r^* x_1(t-r) + B_0^* e(t) + \ldots + B_{r-1}^* e(t-r+1).$$

Next, partition $x_1(t)$ conformably with A_k^* and B_k^* in (2.2), as $x_1(t) = [x_{11}(t)^T, x_{12}(t)^T, x_{13}(t)^T]^T$, where $x_{11}(t)$ is $n_1 \times 1$ and $x_{12}(t)$ and $x_{13}(t)$ are $n_2 \times 1$, so that, using $A_k = 0$ for $k > p$ and $B_k = 0$ for $k > q$, (2.5) becomes

$$(2.6) \qquad [x_{11}(t)^T, x_{12}(t)^T]^T = A_1[x_{11}(t-1)^T, x_{12}(t-1)^T]^T + \ldots + A_p[x_{11}(t-p)^T, x_{12}(t-p)^T]^T + B_0 e(t) + \ldots + B_q e(t-q),$$

$$(2.7) \qquad x_{13}(t) = C_1 x_{12}(t-1) + \ldots + C_\upsilon x_{12}(t-\upsilon).$$

So far, we have not set $[x_{11}(t)^T, x_{12}(t)^T]$. Letting $[x_{11}(t)^T, x_{12}(t)^T]^T = [u_1(t)^T, u_2(t)^T]^T = u(t)$, for all t, we see that (2.1) and (2.6) are identical. Therefore, (2.3) is a valid state-space representation of (2.1).

With $x_{12}(t) = u_2(t)$, we see that $x_{13}(t)$ is the "lagged" part in temporally agreggated or delayed observations. This state-space

representation extends a common one (e.g., Ansley and Kohn 1983) to include temporal aggregation. The meaning of $x_k(t)$, for $k = 2, \ldots, r$, now becomes clear: these state variables cumulate AR terms, MA terms, and lagged observations in a compact way.

The observation equation is constructed in two steps. Let $\zeta(t)$ be an $m \times 1$, unobserved vector of observation errors which is partitioned conformably with $w(t)$ as $\zeta(t) = [\zeta_1(t)^T, \zeta_2(t)^T]^T$. Then, in the absence of cross-sectional aggregation, error-corrupted stocks and flows are, respectively, observed as $w_1(t) = x_{11}(t) + \zeta_1(t)$ and $w_2(t) = C_0 x_{12}(t) + x_{13}(t) + \zeta_2(t)$, or, equivalently, as

$$(2.8) \qquad w(t) = \Delta x(t) + \zeta(t),$$

where $$\Delta = \begin{bmatrix} I_{n_1} & 0 & 0 & 0 \\ 0 & C_0 & I_{n_2} & 0 \end{bmatrix},$$

I_{n_1} is the $n_1 \times n_1$, identity matrix, and I_{n_2} is the $n_2 \times n_2$, identity matrix. If some stocks or flows are observed as cross-sectional aggregates, then, Δ is appropriately reduced by row-wise summations.

Following usual practice, we assume that $\zeta(t)$ is a Gaussian, white noise, with zero mean and a constant, covariance matrix, and that it is uncorrelated with all values of the state vector and its disturbances. That is, $\zeta(t) \sim \text{NIID}[0, \Sigma_\zeta]$, where $\Sigma_\zeta \geq 0$ (positive semi-definite), and $E\zeta(\tau)e(t)^T = 0$ and $E\zeta(\tau)x(1)^T = 0$, for all τ, $t \geq 1$. As in the effective reparameterization of Σ_e to B_0 = lower triangular with the normalization $\Sigma_e = I_n$, it is convenient to reparameterize Σ_ζ to R, where R is lower triangular and satisfies $RR^T = \Sigma_\zeta$, and, thereby, automatically impose $\Sigma_\zeta \geq 0$.

So far, Δ has been described as a matrix of zeroes and ones which selects and, cross-sectionally and temporally, aggregates state variables. However, Δ may also incorporate contemporaneous interactions among observable variables (a simultaneous-equations part of the model), which cannot be put into the law of motion, (2.3), at least if the state is to be of minimal dimension. Such a part of Δ will depend on the parameter vector, ϕ. For example, this will occur in a linear rational expectations model with explicit dynamic optimization, if one of the equations in (2.8) represents an agent's optimal linear decision rule (e.g., Hansen and Sargent 1980).

To conclude the construction of the observation equation, let $y(t)$ denote the $m(t) \times 1$ vector of values of w for period t which are actually observed, where $m(t) \leq m$. Therefore, we have $y(t) = \Lambda(t)w(t)$, where $\Lambda(t)$ is the $m(t) \times m$ selection matrix which picks out the observed elements of $w(t)$. Combining (2.8) and $y(t) = \Lambda(t)w(t)$, we get the observation equation

$$y(t) = D(t)x(t) + v(t), \tag{2.9}$$

where $D(t) = \Lambda(t)\Delta$ and $v(t) = \Lambda(t)\zeta(t)$. Disturbance $v(t)$ inherits the properties of $\zeta(t)$: $v(t) \sim \text{NIID}[0, \Sigma_v(t)]$, where $\Sigma_v(t) = \Lambda(t)\Sigma_\zeta\Lambda(t)^T$ and $Ev(\tau)e(t)^T = 0$, for all τ and t.

In the application, $n = 2$ and $u(t) = [u_1(t), u_2(t)]^T = [\%\Delta EMP(t), \%\Delta GNP(t)]^T$, where $\%\Delta EMP$ and $\%\Delta GNP$ are annualized, percentage changes in total employment and real GNP (the data and their transformation to percentage changes are discussed in detail in section 4). Both variables are treated as observed without delay. $\%\Delta EMP$ is treated as a stock and

%ΔGNP is treated as a flow, as a three-month sum. Therefore, $w(t) = [w_1(t), w_2(t)]^T$, where $w_1(t) = u_1(t)$ and $w_2(t) = u_2(t) + u_2(t-1) + u_2(t-2)$, and $\Delta = \begin{bmatrix} 1 & 0 & 0 & 0 & \dots & 0 \\ 0 & 1 & 1 & 0 & \dots & 0 \end{bmatrix}$, where Δ has a total of $3 \cdot \max(p, q+1, 2)$ columns. %ΔEMP is observed every month, for $t = 1, \dots, N$. Observations on %ΔGNP are registered in the third month of a quarter. Suppose, for the sake of illustration, that $t = 1$ is the first month of a quarter (e.g., January). Then, when %ΔEMP and %ΔGNP are both observed, for $t = 3, 6, \dots, m(t) = 2$ and $\Lambda(t) = I_2$, the 2×2 identity matrix; when only %ΔEMP is observed, for $t = 1, 2, 4, 5, \dots, m(t) = 1$ and $\Lambda(t)$ equals the 1×2 vector $[1, 0]$.

The idea of accounting for missing data in this way, by zeroing out relevant state variables with a time-varying observation equation, originates with Jones (1980) and was extended to multivariate cases by Ansley and Kohn (1983). The idea of treating mixed-frequency data in this way was proposed by Zadrozny (1988).

Although, following the application and usual practice, we have assumed that the model, (2.1), and the first part of the observation equation, (2.8), are time invariant, the likelihood computations of section 3 remain valid if (2.1) and (2.8) are time varying in the sense that their coefficients are known, nonstochastic functions of time. When this is the case, then, one simply has to add time arguments to the coefficients, i.e., write $A_0(t), \dots, B_q(t), \Sigma_e(t), \Delta(t)$, and $\Sigma_\zeta(t)$. For example: $A_0, \dots, B_q, \Sigma_e$ will be time varying when (2.1) is a period-by-period linearization of a nonlinear model; Δ will be time varying when (2.8) is the regression equation in a time-varying regression model (e.g., Judge et al. 1980, pp. 391-397); and, Σ_ζ will be time varying when data come from sample surveys of different sizes and accuracies.

3. Computing the Likelihood Function with the Kalman Filter.

Let $\theta = [\text{vec}(A_1)^T, \ldots, \text{vec}(A_p)^T, \text{vech}(B_0)^T, \text{vec}(B_1)^T, \ldots, \text{vec}(B_q)^T]^T$ collect the unnormalized coefficients of (2.1), which depend on structural parameters to be estimated, where vec(•) vectorizes a matrix columnwise and vech(•) vectorizes the lower-triangular part of a matrix columnwise. Σ_ζ (or its reparameterization R) is not included in θ, because B_0, which is the effective reparameterization of Σ_e, and Σ_ζ are not both identifiable and, following usual procedure, we propose setting Σ_ζ according to the presumed accuracy of the data and estimating B_0.

A particular model is characterized by a mapping, $\theta = \Psi(\phi)$, from ϕ to θ. For computational purposes, we treat ϕ as lying in an unrestricted set (Euclidian space), so that any equality or strict inequality restrictions are built into Ψ. Non-strict inequality restrictions can be accounted for by attaching a penalty to the likelihood function. We propose to estimate (2.1) by maximizing the Gaussian likelihood with respect to ϕ, under the restrictions in Ψ. Under general smoothness restrictions on Ψ, estimates will be consistent and asymptotically efficient and will have a Gaussian, asymptotic distribution. (To obtain the asymptotic properties, in theory, one may have to restrict ϕ to a compact region.) Within limits, these properties are preserved when the model and the assumption of a Gaussian distribution differ from the data generating process. See, for example, Hannan (1979) and the references therein.

In this paper, we shall simply take for granted restrictions which guarantee these properties of the estimates and shall instead

concentrate on restrictions on parameters which ensure that the likelihood function is well-defined. As we discuss further in this section, the likelihood function is well-defined when the model implies that the data have a nonsingular, joint, probability distribution.

Let $Y(t) = \{y(1), \ldots, y(t)\}$, for $t = 1, \ldots, N$, so that $Y(N)$ is the full sample of observations on $y(t)$. Let $L(t)$, for $t = 1, \ldots, N$, denote the nonconstant part of -2 times the exact, Gaussian, log-likelihood function of $Y(t)$. Although $L(t)$ is directly a function of θ and a composite function of ϕ, for simplicity, we suppress reference to θ and ϕ. Let $x(t|t-1) = E[x(t)|Y(t-1)]$ and $y(t|t-1) = E[y(t)|Y(t-1)]$, with associated errors, $\tilde{x}(t) = x(t) - x(t|t-1)$ and $\xi(t) = y(t) - y(t|t-1)$, and error covariances, $V(t) = E[\tilde{x}(t)\tilde{x}(t)^T]$ and $M(t) = E[\xi(t)\xi(t)^T]$. Vector $\xi(t)$ is called the innovation of $y(t)$, because it represents new information in $y(t)$, beyond that contained in $Y(t-1)$.

The Kalman filter (KF) provides one way of decomposing a time series into its innovations. In fact, the KF itself has many possible implementations. A comprehensive discussion of theoretical and computational properties of various KF implementations is given by Anderson and Moore (1979). Schweppe (1965) is usually cited as the first one to propose computing the Gaussian likelihood function of a time series with its innovations.

The following equations are one possible implementation of the KF which, given the model, its parameters, and the data, computes $L(N)$ by iterating over the sampling times, $t = 1, \ldots, N$. At the start of iteration t, $x(t|t-1)$, $V(t)$, and $L(t-1)$ are given from the previous iteration. Given the values of these quantities, $L(t-1)$ is updated with

$$M(t) = \Sigma_v(t) + D(t)V(t)D(t)^T, \tag{3.1}$$

$$\xi(t) = y(t) - D(t)x(t|t-1), \tag{3.2}$$

$$L(t) = L(t-1) + \ln|M(t)| + \xi(t)^T M(t)^{-1}\xi(t), \tag{3.3}$$

where $|\cdot|$ denotes the determinant; $x(t|t-1)$ and $V(t)$ are updated with

$$K(t) = FV(t)D(t)^T M(t)^{-1}, \tag{3.4}$$

$$x(t+1|t) = Fx(t|t-1) + K(t)\xi(t), \tag{3.5}$$

$$\Phi(t) = F - K(t)D(t), \tag{3.6}$$

$$V(t+1) = GG^T + FV(t)\Phi(t)^T. \tag{3.7}$$

Equations (3.1) to (3.7) are written in the order in which computations proceed. Equations (3.1), (3.2), (3.4), and (3.6) are essentially definitions of $M(t)$, $\xi(t)$, $K(t)$, and $\Phi(t)$. $K(t)$ is called the Kalman gain matrix. $\Phi(t)$ is called the closed-loop matrix. Equation (3.7) is a version of the Riccati equation. As we discuss in the appendix, under certain conditions (3.1) to (3.7) are a numerically stable implementation of the KF.

To start the iterations, $x(1|0)$ and $V(1)$ must be specified, and, of course, $L(0) = 0$. We limit the discussion on how to set $x(1|0)$ and $V(1)$ to the stationary case. A model is stationary when its law of motion and the structural part of its observation equation are time

invariant and when its state law of motion is asymptotically stable. The latter condtition means that F is a (discrete-time) stable matrix, i.e., has all eigenvalues less than one in modulus. Ansley and Kohn (1985) and Bell and Hillmer (1987) describe a data-transformation method for initializing $x(1|0)$ and $V(1)$, which, although designed for the nonstationary case, also applies in the stationary case.

The exact, likelihood function is obtained in the stationary case when $x(1|0) = \mu_x$, the unconditional mean of x, and $V(1) = \Sigma_x$, the unconditional covariance of x. In the stationary case, when the data have been adjusted for means and fixed effects, $\mu_x = 0$ and Σ_x solves the (discrete-time, algebraic) Lyapunov equation,

$$\Sigma_x - F\Sigma_x F^T = GG^T. \tag{3.8}$$

When F is a stable matrix, (3.8) yields a unique, symmetric, positive semi-definite value of Σ_x (Anderson and Moore 1979, pp. 64-67). When, in addition, the state vector is controllable or stabilizable, then, $\Sigma_x > 0$ (controllability is defined in the appendix).

In sum, in the stationary case, the exact likelihood function is obtained when $x(1|0) = 0$ and $V(1) = \Sigma_x$, where Σ_x solves (3.8).

There are various methods for solving (3.8). An obvious, but inefficient, way to solve (3.8) is to apply the vectorization rule $vec(ABC) = [C^T \otimes A]\cdot vec(B)$ (Neudecker 1969, p. 954), where A, B, and C are conformable matrices and $\otimes$ is the Kronecker product, to obtain

$$\{I_{s^2} - [F \otimes F]\}vec(\Sigma_x) = vec(GG^T), \tag{3.9}$$

where I_{s^2} is the $s^2 \times s^2$ identity matrix, s being the dimension of x(t). Equation (3.9) is in the standard form of a linear system, Ab = c, and can, therefore, be solved for $b = \text{vec}(\Sigma_x)$ by any standard method (Golub and Van Loan 1983, pp. 52-80). Kohn and Ansley (1982) describe how to exploit the symmetry of Σ_x and GG^T and the companion form of F in the solution of (3.9). In the application, we exploited symmetry, but did not attempt to exploit the companion form of F.

Next, we state restrictions on parameters which guarantee that the likelihood function is well-defined, hence, computable. L(N) is well-defined if the model and the sampling scheme transmit enough variations from e and ζ to y so that $\{y(t)\}_{t=1}^N$ have a nonsingular, joint, probability distribution. Because, given $x(1|0)$ and $V(1) \geq 0$, the KF effects a one-to-one transformation from $\{y(t)\}_{t=1}^N$ to $\{\xi(t)\}_{t=1}^N$, $\{y(t)\}_{t=1}^N$ has a nonsingular distribution if and only if $M(t) = E\xi(t)\xi(t)^T > 0$, for $t = 1, \ldots, N$. If $V(1) \geq 0$ and $\Sigma_v(t) + D(t)GG^TD(t)^T > 0$, for $t = 1, \ldots, N$, then, a simple induction proof shows that $M(t) > 0$, for $t = 1, \ldots, N$. In the stationary case, (3.8) yields $V(1) = \Sigma_x \geq 0$. Therefore, L(N) is computable if $\Sigma_v(t) = \Lambda(t)\Sigma_\zeta\Lambda(t)^T > 0$, if $D(t)GG^TD(t)^T > 0$, or if both of these conditions hold, for $t = 1, \ldots, N$. In practice, GG^T usually has much less than full rank, but D(t) has full row rank and picks a full-rank part out of GG^T, so that $D(t)GG^TD(t)^T > 0$, for $t = 1, \ldots, N$. In the application, $\Sigma_\zeta > 0$ and B_0 has full column rank, conditions which, given the sampling scheme discussed at the end of section 2, respectively, imply $\Sigma_v(t) > 0$ and $D(t)GG^TD(t)^T > 0$, hence, $M(t) > 0$, for $t = 1, \ldots, N$.

We conclude this section by recommending some efficiencies in computations with (3.1) to (3.9). Whenever possible, known blocks of

zeroes (sparsity) and ones in the given matrices, F, G, D(t), and $\Sigma_v(t)$, should be exploited to avoid unnecessary arithmetic. The computed matrices, M(t), K(t), $\Phi(t)$, and V(t), are generally dense (not sparse), even when the given matrices are sparse. An exception is that $\Phi(t)$ inherits the companion form of F when D(t) only picks linear combinations out of the first $n_* \times 1$ block of x(t), as in the present setup. In our experience, unless sparsity and KF convergence are exploited, applications at least as complex as the present one are intractable on a personal computer.

Also, rather than invert M(t) and multiply with its inverse in (3.3) and (3.4), it is better to first Cholesky factorize M(t) and, then, solve the appropriate triangular, linear systems. When $M(t) > 0$, it has the unique, Cholesky factorization $M(t) = \Omega(t)\Omega(t)^T$, where $\Omega(t)$ is lower triangular, with positive elements on the principal diagonal. For example, in (3.3), it is recommended to first solve $\Omega(t)\eta(t) = \xi(t)$ by forward elimination and, then, compute $\xi(t)^T M(t)^{-1}\xi(t)$ as $\eta(t)^T\eta(t)$. See, e.g., Golub and Van Loan (1983, pp. 52-54 and 86-92). In addition, Cholesky factorization of M(t) provides an on-line test of whether the computations are proceeding successfully. The full set of computations is successful when $M(t) > 0$, for $t = 1, \ldots, N$, and $M(t) > 0$ if and only if $\Omega_{ii}(t) > 0$, for $i = 1, \ldots, m(t)$. In the Cholesky factorization algorithm, $\Omega_{ii}(t)$ is obtained by taking the square root of a positive number. The algorithm fails at subiteration (i,t) when this number is not positive: $\Omega_{ii}(t) > 0$ cannot be computed, so that M(t) is not positive definite. Finally, we note that it is best to compute $\ln|M(t)|$ as $2 \cdot \sum_{i=1}^{m(t)} \ln[\Omega_{ii}(t)]$.

4. Application to Forecasting U.S. GNP at Monthly Intervals.

The data used in the application were obtained in April 1989 from the Federal Reserve Board. They span January 1947 to December 1988 and are seasonally adjusted. GNP is on a 1982 basis. The data are historical series, which mix revised, early values with preliminary, recent values. Although we took account of observation errors, we treated the data as being homogeneous in their degree of revision. This is justified because the data mostly comprise final revisions. The employment data are produced and published, in Employment and Earnings, by the Bureau of Labor Statistics. They are based on monthly surveys of payrolls of establishments and represent totals over all types of employees and establishments. The GNP data are produced and published, in Survey of Current Business, by the Bureau of Economic Analysis. They are based on numerous surveys and are produced and revised by a lengthy process (e.g., Carson 1987, Young 1987).

The given data are $Y_1(t) = EMP(t)$, for $t = 1$ (January 1947), 2, 3, ..., where EMP(t) denotes the stock of employment in month t, and $Y_2(t) = GNP(t) + GNP(t-1) + GNP(t-2)$, for $t = 3, 6, 9, \ldots$, where GNP(t) denotes the (unobserved) accumulated flow of real GNP in month t. Prior to using them, we transformed the data into mean-adjusted, (approximate) annualized, percentage changes as $y_1(t) = \%\Delta EMP(t) - \overline{\%\Delta EMP}$, for $t = 2, 3, 4, \ldots$, and $y_2(t) = \%\Delta GNP(t) - \overline{\%\Delta GNP}$, for $t = 6, 9, 12, \ldots$, where $\%\Delta EMP(t) = 1200\{\ln[Y_1(t)] - \ln[Y_1(t-1)]\}$ and $\%\Delta GNP(t) = 400\{\ln[Y_2(t)] - \ln[Y_2(t-3)]\}$. To call EMP and GNP stocks and flows is a purely interpretive statement, to clarify the meaning of the given data. By contrast, as we now explain, for a given (structural) model of the underlying variables, (2.1), each classification of %ΔEMP and %ΔGNP into

stocks and flows yields an effectively different (reduced form) model of the observations and, therefore, has substantive implications for estimation and forecasting.

Momentarily write $y_2(t)$ as $y_2^q(t)$ and define $y_2^m(t)$ as $400\{\ln[Y_2(t)] - \ln[Y_2(t-1)]\}$ adjusted for mean, so that $y_2^q(t) = y_2^m(t) + y_2^m(t-1) + y_2^m(t-2)$. Now, in the notation of section 2, $y_1(t)$ was treated as a stock in the sense that $y_1(t) = w_1(t) + \zeta_1(t)$, where $w_1(t)$ is the true value of $y_1(t)$, and $y_2(t)$ was treated as a flow in the sense that $y_2(t) = w_2(t) + w_2(t-1) + w_2(t-2) + \zeta_2(t)$, where $w_2(t)$ is the true value of $y_2^m(t)$. (As we noted in section 2, $y_2(t)$ is registered in the third month of a quarter.) By contrast, if $y_2(t)$ were treated as a stock, then, we would have $y_2(t) = w_2(t) + \zeta_2(t)$, where $w_2(t)$ would be the true value of $y_2^q(t)$.

The difference is that in the flow case $y_2(t)$ enters the model as the true value of $y_2^m(t)$, whereas in the stock case $y_2(t)$ enters the model as the true value of $y_2^q(t)$. Treating %ΔGNP as a flow is more appealing because it places %ΔGNP in the model symmetrically with %ΔEMP, as a monthly first difference. Generally, treating a variable as a flow implicitly imposes MA-type restrictions on disturbances of the model, so that misspecifying a variable as a flow generally reduces the fit and forecasting ability of a model. As it was, treating %ΔGNP as a flow markedly improved fit and forecasting ability.

We transformed the data into annualized, percentage changes in order to make the forecasts comparable with those of others and to make the data stationary. In particular, by removing trends, differencing made the data stationary. Because trends dominated variations in the undifferenced data, a failure to remove them would have resulted in

severe collinearity problems in estimation and biased tests. Because the transformed data display somewhat greater variations from 1947 to 1957 than in later years, to be sure that we had a stationary sample, we only used values from 1958 to 1988.

There has been some discussion lately about whether trends should be removed by differencing or by linear regression (e.g., Whiteman and Roberds 1989). A generalization of differencing, which recognizes that variables may share common trends, is to treat them as cointegrated (Engle and Granger 1987). The approach of Fountis and Dickey (1989) for isolating common trends among cointegrated variables is a natural one in a state-space framework. The model can also legitimately be estimated with undetrended data if appropriate restrictions are placed on the parameters. This has been done especially in Bayesian analyses (e.g., Doan, Litterman, Sims 1984, Roberds 1988).

We now discuss how we set Σ_ζ, the covariance matrix of observation errors. Let $Y_i^*(t)$ and $y_i^*(t)$, for $i = 1$ and 2, denote true values corresponding to observations, $Y_i(t)$ and $y_i(t)$. (Note that $y_i^*(t)$ is equivalent to $w_i(t)$ of section 2.) Because the model is specified in terms of the transformed data, we set $\zeta_i(t) = y_i(t) - y_i^*(t)$. Because $\ln[Y_i(t)] - \ln[Y_i^*(t)] \cong \{Y_i(t) - Y_i^*(t)\} \div Y_i^*(t)$, we define the relative error of observing $Y_i(t)$ as $\varepsilon_i(t) = \ln[Y_i(t)] - \ln[Y_i^*(t)]$. Let $\mu_i = E\{\ln[Y_i(t)] - \ln[Y_i(t-b_i)]\}$, $\mu_i^* = E\{\ln[Y_i^*(t)] - \ln[Y_i^*(t-b_i)]\}$, and $\delta_i = \mu_i - \mu_i^*$. Then, using $y_i(t) = a_i\{\ln[Y_i(t)] - \ln[Y_i(t-b_i)] - \mu_i\}$ and $y_i^*(t) = a_i\{\ln[Y_i^*(t)] - \ln[Y_i^*(t-b_i)] - \mu_i^*\}$, where $[a_1, a_2] = [1200, 400]$ and $[b_1, b_2] = [1, 3]$, we obtain $\zeta_i(t) = a_i\omega_i(t)$, where $\omega_i(t) = \varepsilon_i(t) - \varepsilon_i(t-b_i) - \delta_i$.

The relation $\omega_i(t) = \varepsilon_i(t) - \varepsilon_i(t-b_i) - \delta_i$ can be viewed in two

ways. Either way we run into a difficulty. First, to be consistent with the assumption that $\zeta_i(t)$ is serially uncorrelated, we can assume that $\omega_i(t)$ is serially uncorrelated. But, then, $\varepsilon_i(t)$ is a random walk, so that, strictly speaking, the available covariance information on $\varepsilon_i(t)$ cannot be used to restrict the covariances of $\omega_i(t)$ and $\zeta_i(t)$. On the other hand, we can assume that $\omega_i(t)$ follows a stationary, ARMA model. But, then, the state-space representation should be extended to incorporate the model of $\omega_i(t)$.

If there are direct measurements of $\zeta_i(t)$, the model of $\omega_i(t)$ can be estimated prior to estimating the principal model, (2.1). For example, $\zeta_2(t)$ of GNP could be measured as the difference between preliminary and revised observations (e.g., 15- and 45-day estimates). We leave this possibility for a future investigation. Without such measurements, an identification problem prevents the model for $\omega_i(t)$ from being estimated simultaneously with (2.1).

We followed the second option, that $\omega_i(t)$ is covariance stationary, but did not attempt to incorporate a model of $\omega_i(t)$ in the state-space representation. That is, we ignored the contradiction between treating $\zeta_i(t)$ as a white noise in the Kalman filter and using serial correlations of $\omega_i(t)$ in setting Σ_ζ. Because δ_i can absorb the mean of ε_i, we set $E\varepsilon_i(t) = 0$ and, consistent with the previous assumption that $E\zeta_i(t) = 0$, we assumed that $\delta_i = 0$.

Because employment and GNP data are constructed with different source data, we assumed that $\mathrm{Cor}[\varepsilon_1(t), \varepsilon_2(t)] = 0$, hence, that $\mathrm{Cor}[\zeta_1(t), \zeta_2(t)] = 0$. Letting $\sigma_i^2 = \mathrm{Var}[\varepsilon_i(t)]$ and interpreting σ_i as the relative, sampling error or relative, statistical discrepancy of a variable, Employment and Earnings and Survey of Current Business tell us

that $\sigma_1 = .001$ and $\sigma_2 = .0015$. Therefore, $\omega_i(t) = \varepsilon_i(t) - \varepsilon_i(t-b_i)$ implies that $\Sigma_\zeta = \text{diag}[2a_1^2\sigma_1^2(1 - \rho_1), 2a_1^2\sigma_1^2(1 - \rho_2)] = \text{diag}[2.88(1 - \rho_1), 0.72(1 - \rho_2)]$, where $\rho_i = \text{Cor}[\varepsilon_i(t), \varepsilon_i(t-b_i)]$. To resolve Σ_ζ, we assumed that $\rho_1 = \rho_2$ and considered the values $\rho_i = -1, -.5, 0, .5$, and .99. Although the results did not vary dramatically over these values of ρ_i, the best results, in terms of lowest serial correlations of innovations, highest t-ratios of parameters, and highest forecasting accuracy, were obtained with $\rho_i = .5$. Therefore, the reported results are all based on $\rho_i = .5$, hence, on $\Sigma_\zeta = \text{diag}[1.44, .36]$.

We searched for the best model of the transformed data, for January 1958 to December 1978, by starting with the ARMA(3,1) model and reducing it in stages to the ARMA(1,1) model. Under general conditions (e.g., Hannan 1979), the estimated parameter vector, $\hat{\theta}$, is asymptotically distributed as $N[0, \Xi]$, where Ξ is consistently estimated by $\hat{\Xi} = 4[\nabla\hat{L}(N)^T \cdot \nabla\hat{L}(N)]^{-1}$ and $\nabla\hat{L}(N)$ = numerically approximated, row-form gradient of $L(N)$ evaluated at $\hat{\theta}$. Therefore, $t_i = \hat{\theta}_i/\hat{s}_i$ is asymptotically distributed as $N[0, 1]$, where $\hat{s}_i$ = positive square-root of the (i,i) element of $\hat{\Xi}$. At each stage, we considered setting to zero parameters which were highly insignificant in the sense that $|\hat{\theta}_i| < .1$ and $|t_i| = |\hat{\theta}_i/\hat{s}_i| < .1$. The insignificant parameters were eliminated only when doing so reduced bias-corrected Akaike information criterion (AIC). The reductions in AIC were always ≥ 1 in absolute value. There is as yet no general means of judging statistical significance of reductions in AIC.

Akaike (1973) introduced AIC as an operational approximation of the Kullback-Leibler measure of information. Minimum AIC guides one to the true data generating process because Kullback-Leibler information is

minimized (at zero) when the model is identical to the data generating process and increases as the model diverges from the data generating process. The theory of AIC has recently been further extended by Findley and Wei (1989). Because the originally specified AIC is biased in finite samples, in the sense that it tends to overfit the data, others have proposed versions of AIC which attempt to correct for this bias by penalizing the number of parameters more heavily. We used the version proposed by Hurvich and Tsai (1989).

Figures 1 to 4 report own- and cross-serial correlations of the transformed data, for January 1958 to December 1988. Figures 1 to 3 report significant, own-serial correlations of employment and significant, cross-serial correlations of employment and GNP, at 0 to 6 month lags. The especially significant, employment-GNP, cross-serial correlations, at 1 to 4 month lags, in Figure 3, show the potential for using employment to forecast GNP. Figure 4 reports insignificant own-serial correlations of GNP.

Table 1 reports estimation statistics of the best model. Considering that the data were adjusted for trend, an R^2 of .228 for employment is fair and an R^2 of .472 for GNP is good. Although R^2, defined as 1 - variance of innovations ÷ variance of observations, is not a reliable statistic (e.g., in the present estimation method it can be < 0), nevertheless, it conveniently summarizes the fit of individual equations in a model. Figures 5 to 8 report own- and cross-serial correlations of innovations of the estimated model. Judging by 2-standard-error confidence bounds on individual correlations and Ljung-Box (1978) Q statistics, the model successfully reduced the data to white-noise innovations. Only $\hat{\theta}_1$, $\hat{\theta}_4$, and $\hat{\theta}_5$ appear to be significant at

conventional 10% and 5% levels ($|t_i| \geq 1.645, 1.96$). Perhaps, t_4 is unusually large because the numerical approximation of $\hat{\nabla L}(N)$ was insuffuciently accurate. Among remaining parameters, only $\hat{\theta}_2$ is in the range of significance ($|t_i| \geq 1$). Setting to zero the least significant parameters, θ_3 and θ_6 to θ_{11}, increased AIC and reduced forecasting ability.

This result and the high correlations among $\hat{\theta}_3$ and $\hat{\theta}_6$ to $\hat{\theta}_{11}$, indicated by $\hat{\Xi}$, suggest that linear restrictions -- but not simple, zero restrictions -- might be imposed on θ_3 and θ_6 to θ_{11}. This could be done with orthogonal, t ratios, as follows. Suppose that $\hat{\Xi}$ has an eigenvalue decomposition, $\hat{\Xi} = Z\Gamma Z^{-1}$, where Γ is the diagonal matrix of positive, real eigenvalues, γ_i, and Z is the matrix of column eigenvectors, z_i. Furthermore, suppose that the eigenvalues are distinct, so that Z is orthogonal ($Z^TZ = I$). Then, $\tau = \Gamma^{-\frac{1}{2}}Z^T\hat{\theta}$ is the vector of orthogonal t ratios, with i-th element $\tau_i = z_i^T\hat{\theta}/\gamma_i^{-\frac{1}{2}}$. That is, τ_i is asymptotically distributed as N[0, 1], independently of τ_j, for $j \neq i$. The idea is to impose the linear restriction $z_i^T\theta = 0$ if τ_i is insignificant. Imposing $z_i^T\theta = 0$ is equivalent to restricting θ along a long axis of an elliptical, confidence region in the parameter space.

To compute forecasts, first note that (2.3), (2.9), and the white-noise assumptions on their disturbances imply that

$$x(t+k+1|t) = Fx(t+k|t), \tag{4.1}$$

$$y(t+k|t) = D(t+k)x(t+k|t), \tag{4.2}$$

for $k = 1, \ldots, \bar{k}$, where $x(t+k|t) = E[x(t+k)|Y(t)]$ and $y(t+k|t) =$

$E[y(t+k)|Y(t)]$. Let $t = 1, \ldots, N_1$ denote the estimation period and let $t = N_1+1, \ldots, N_2$ denote the forecasting period. Set $\theta = \hat{\theta}$, Σ_ζ as prescribed, $x(1|0) = 0$, and $V(1)$ with (3.8). Given these values, iterate with the Kalman filter over $t = 1, \ldots, N_1-\bar{k}$, to obtain $x(N_1-\bar{k}+1|N_1-\bar{k})$. Given $x(N_1-\bar{k}+1|N_1-\bar{k})$, iterate with (4.1) and (4.2), for $k = 1, \ldots, \bar{k}$. Using (3.5), update $x(N_1-k+1|N_1-k)$ to $x(N_1-\bar{k}+2|N_1-\bar{k}+1)$. Given $x(N_1-\bar{k}+2|N_1-\bar{k}+1)$, iterate with (4.1) and (4.2), for $k = 1, \ldots, \bar{k}$. Continue in this fashion, for $t = N_1-\bar{k}+2, \ldots, N_2-1$, to obtain the desired forecasts.

Note also that all data were adjusted with estimation-period sample means. This made the forecasting exercise more realistic. If estimation and forecasting period means are different and data in the forecasting period are adjusted with their own means, then, in effect, the forecasts are based on advance knowledge of mean shifts.

Table 2 reports standard errors of forecasts of the best model over the period January 1979 to December 1988. Table 3 compares standard errors of the GNP forecasts with ones for similar periods reported by McNees (1986), Roberds (1988), and Trehan (1989). Considering that they are based on a smaller information set (observations on fewer variables), the present forecasts compare favorably. Table 2 also reports Theil U statistics, defined as U = standard error of a forecast ÷ standard error of the corresponding naive forecast, where the naive forecast is the most recent observation at the time the forecast is made. Theil U statistics are free of units of measurement and provide a basic test of the competitiveness of a set of forecasts. If forecasts which cost something to be produced yield U's $\geq$ 1, then, they are dominated by the costless, naive method and are uncompetitive. In this

respect, present forecasts are quite competitive.

To see how much cross-variable feedbacks contributed to the forecasts, we developed univariate forecasts which are reported in Tables 4 and 5. In both bases, AR(1) models were judged to be the best models. Comparing Tables 4 and 5 with Table 2, we see that GNP contributes marginally to forecasting employment, but that employment contributes substantially to forecasting GNP. In the latter case, the quarterly statistics are compared with monthly statistics for 3, 6, 9, and 12 months ahead.

To see how well the present method exploited mixed-frequency, sample information, we developed forecasts with a bivariate, AR model restricted to mimic the limited ability of conventional, regression methods to use mixed-frequency data. In this respect, GNP was restricted to quarterly lags. We started with an AR(3) model and, using minimum AIC, reduced it to an AR(1) model. Estimation and forecasting statistics of the model are reported in Tables 6 and 7. Compared with the unrestricted, ARMA(1,1) model in Table 1, the restricted, AR(1) model yielded a similar R^2 of .227 for employment, a notably lower R^2 of .304 for GNP, and mostly significant t ratios. Except for GNP-employment, cross-serial correlations, which were on the borderline of significance, the model displayed insignificant serial correlations of innovations. In particular, GNP-employment, cross-serial correlations at 2 and 10 month lags had significant values of .318 and -.284. Evidently, the present method was much more successful at capturing feedbacks from employment to GNP, even though conventional, regression methods allow employment to appear in the GNP equation at any monthly lags. Comparing Tables 2 and 7, we see that this advantage translated into uniformly better forecasts

of both variables, in particular, substantially better forecasts of GNP from 1 to 5 months ahead.

The computations were carried out with a FORTRAN program written by the author. In the program, the likelihood function is maximized with a trust-region method, using MINPACK subprograms (More' et al. 1980), and many of the matrix computations are done with EISPACK (Smith et al. 1976) and LINPACK (Dongarra et al. 1979) subprograms. The program was compiled with version 3.00 of the Lahey Computer Systems FORTRAN 77 compiler and was run on a Northgate Computer Systems 386 personal computer, operating at 20 MHz clock speed with a numerical co-processor. Single-precision arithmetic (about 6 significant decimal digits) was used. In estimating the best model, each iteration took about 26 seconds. Computing forecasts of the best model took about 1 minute and 20 seconds. Doing all of the computations underlying the reported tables and figures took about 25 minutes.

5. Conclusion.

This paper has developed and applied a Kalman-filtering method for estimating a multivariate ARMA model with mixed-frequency, stock-flow data. The application shows that the method can usefully exploit cross-frequency, sample information and is computationally tractable. Virtually any attempt to estimate a multivariate, economic, time-series model confronts the problem of efficiently using mixed-frequency, stock-flow data. The present method efficiently solves this problem. The method allows one to easily account for the sampling complications of mixed frequencies and other patterns of missing data, temporal aggregation (stocks versus flows), cross-sectional aggregation,

precision of observations, and timeliness of observations. In particular, because one does not have the problem of reformulating the model to conform with the sampling scheme, the method allows one to use the available sample information, without compromising either the model or the sampling scheme.

The best model which we obtained adequately fit data for 1958 to 1978 and yielded superior short-term and competitive medium-term forecasts for 1979 to 1988. Although many parameters -- especially MA parameters -- were individually insignificant, judging by AIC and forecasting ability, they were jointly significant. This suggests that by imposing appropriate restrictions, individual significance of parameters could be increased, without reducing the fit of the model and, thereby, possibly increasing its forecasting ability. In this respect, tests with orthogonal, t ratios, discussed at the end of in section 4, might suggest useful, linear restrictions. One might also consider restrictions of the form commonly imposed in Bayesian, AR models (e.g., Doan, Litterman, Sims 1984, Roberds 1989). Also, as we mentioned in section 1, one should account for different precisions and timeliness of preliminary and revised data and one should consider other monthly variables to aid in forecasting GNP, e.g., sales, industrial production (see Trehan 1989), and total hours worked = average hours worked per employee × total employment.

Parameters must be identified in order for their estimates to be consistent and asymptotically normally distributed. The best model appeared to be identified locally, because $\hat{\Xi}(N)$ was positive definite, and globally, because different starting values in estimation led to the same estimates. In particular, the model appeared to be identified with

respect to aliasing (e.g., Hansen and Sargent 1983, Christiano 1985). Except for some specialized results (e.g., Christiano 1985), known conditions for identifying ARMA models mostly pertain to single-frequency, stock data (e.g., Hannan 1979 and references therein) and, therefore, provide little information about conditions for identifying ARMA models with mixed-frequency, stock-flow data. The known results do show the close link between identifiability and reconstructibility-controllability (e.g., Burmeister, Wall, and Hamilton 1986). The end of the appendix shows the greatly increased complexity of conditions for reconstructibility-controllability, in the bivariate, ARMA(1,1) model, when single-frequency, stock data is replaced by mixed-frequency, stock-flow data. Conditions for identifying general, ARMA models with mixed-frequency, stock-flow data can be expected to be correspondingly complex.

Appendix. Convergence and Numerical Stability of the Kalman Filter.

Computations with (3.1) to (3.7) are generally intractable unless the KF converges and its convergence is exploited. The KF is said to converge if $M(t)$, $K(t)$, $\Phi(t)$, and $V(t)$ converge to constant values or to periodic cycles as $t \to \infty$. When the KF converges, computations are considerably simplified by skipping (3.1), (3.4), (3.6), and (3.7) and using the limiting values of $M(t)$, $K(t)$, $\Phi(t)$, and $V(t)$. Equations (3.1) to (3.7) indicate that $M(t)$, $K(t)$, and $\Phi(t)$ converge to constant values or to periodic cycles according to whether $V(t)$ does. $V(t)$ converges to a periodic cycle of periodicity ρ if, for any $\varepsilon > 0$, there is a T such that $t \geq T$ implies $\|V(t) - V(t-\rho)\| < \varepsilon$, where $\|\cdot\|$ is some matrix norm. In such case, $\{V(t)\}_{t=1}^{\infty}$ has ρ convergent subsequences. That is, for i =

1, ..., ρ, $\{V(\rho t+i)\}_{t=0}^{\infty}$ converges to $\bar{V}_i$, where $\{\bar{V}_i\}_{i=1}^{\rho}$ constitutes the limiting, periodic cycle. Of course, if $\rho = 1$, $M(t)$, $K(t)$, $\Phi(t)$, and $V(t)$ converge to constant values.

Unless the parameters satisfy certain conditions, (3.1) to (3.7) are numerically unstable. That is, accumulated errors, arising from truncation (or rounding) inherent to finite-precision, computer arithmetic, eventually dominate computed values, often after a relatively small number of iterations. When this happens, the algorithm almost certainly fails, i.e., produces an $M(t)$ which is not positive definite. Verhaegen and Van Dooren (1986) showed that (3.1) to (3.7) are numerically stable if $M(t)$ is sufficiently positive definite ($M(t) \gg 0$), the KF is asymptotically stable, and $V(t)$ is symmetrized each time after it is computed with (3.7). From (3.1), $M(t) \gg 0$ if $\Sigma_v(t) \gg 0$ or $D(t)V(t)D(t)^T \gg 0$. In practice, controllability (defined below) can be counted on to make $D(t)V(t)D(t)^T > 0$ but not necessarily $\gg 0$. Therefore, in practice, one may need $\Sigma_\zeta \gg 0$, hence $\Sigma_v(t) \gg 0$, to ensure $M(t) \gg 0$. Instead of symmetrizing $V(t)$ each time after computing it with (3.7), one can reduce (3.7) so that it only involves the lower- or upper-triangular part of $V(t)$, or, one can use a symmetrical, but computationally more complex, version of (3.7). Our experience confirms Verhaegen and Van Dooren's results: to keep the algorithm numerically stable, in particular, Σ_ζ must be $\gg 0$ and the KF must be asymptotically stable. An increase in computational precision cannot make the algorithm stable; it can only delay the iteration at which the algorithm fails.

In this appendix, we discuss restrictions on parameters which ensure that the KF converges and is asymptotically stable. First, we adapt standard proofs to show that, if the model and sampling exhibit

periodic variation and the state vector is reconstructible and controllable, then, the KF converges to a periodic cycle and is asymptotically stable. These conclusions continue to hold if reconstructibility and controllability are, respectively, weakened to detectability and stabilizability. For brevity, we only explicitly consider the stronger conditions. Second, we discuss restrictions on the parameters of the bivariate, ARMA(1,1) model which ensure that the state vector is controllable and effectively reconstructible (we will explain the qualification effectively). In the application, the sampling scheme has periodicity $\rho = 3$ months and the best, estimated model is a bivariate, ARMA(1,1) model (see Table 1).

To define reconstructibility, let $\mathbf{R}(t) = [D(1)^T, F^T D(2)^T, \ldots, (F^{t-1})^T D(t)^T]^T$. The state vector, x, is said to be (completely) reconstructible if there is a t_r such that $\mathbf{R}(t_r)$ has full (column) rank s, where s is the dimension of x. This property is so called because, abstracting from disturbances and observation errors, there is a t_r at which x can be exactly computed with the accumulated observations, $y(1), \ldots, y(t_r)$. If D(t) is time invariant and x is reconstructible, there will be a $t_r \leq s$, so that there is no need to look for t_r beyond s. This follows from the Cayley-Hamilton theorem, which says that every square matrix satisfies its own characteristic equation, so that the rows of F^t, for $t \geq s$, are linearly dependent on the rows of $F^{s-1}, F^{s-2}, \ldots, F$. Therefore, given that D(t) is time invariant, x is reconstructible if R(s), called the reconstructibility matrix, has full rank s. It is difficult to determine such a general, upper bound, beyond which one need not look for t_r, if D(t) may exhibit any possible, periodic variation.

To define controllability, let $C(t) = [G, FG, \ldots, F^{t-1}G]$. The state vector, x, is said to be (completely) controllable if there is a t_c such that $C(t_c)$ has full (row) rank s. This property is so called because, when e(t) is viewed as a control vector and x_1 and x_* are any initial and target values of x, there is a finite control sequence, $e(1), \ldots, e(t_c)$, which moves x from $x(1) = x_1$ to $x(t_c) = x_*$. Symmetrically with reconstructibility, x is controllable if C(s), called the controllability matrix, has full rank s (in this paper, G is always time invariant).

For a full discussion of reconstructiblity, controllability, and their less restrictive counterparts, detectability and stabilizability, see, e.g., Kwakernaak and Sivan (1972, pp. 53-81 and 459-466).

To emphasize their positions in calender time, we now index the sampling times as $t = t_1, \ldots, t_N$. So far we have, in effect, treated the starting time, t_1, as fixed and the finishing time, t_N, as variable. Now, we consider t_N to be fixed at some value and let $t_1 \to -\infty$. Let Q be any symmetric, positive semi-definite matrix. For each t_1, let $\{V(t_j)\}_{j=1}^{N}$, generated by (3.1), (3.4), (3.6), and (3.7), start with $V(t_1) = Q$. Let $\bar{V}(t_1, t_N)$ denote $V(t_N)$ in $\{V(t_j)\}_{j=1}^{N}$, i.e., the last value in the sequence of V's from t_1 to t_N.

We assume that: (a) $\Sigma_v(t) > 0$, for all t; (b) x is reconstructible, for some (hence, all) t_1 sufficiently below t_N; and, (c) x is controllable. Then, $\bar{V}(-\infty, t_N) = \lim_{t_1 \to -\infty} \bar{V}(t_1, t_N)$ exists. That is, for any $\varepsilon > 0$, there is a T such that $t \leq T$ implies $\|V(t, t_N) - \bar{V}(-\infty, t_N)\| < \varepsilon$. The proof of this result follows by noting that the KF has a dual, linear, optimal, regulator problem in which the analogous result holds. See, e.g., Kwakernaak and Sivan (1972, pp. 490-501 and 525-536). Also,

under assumptions (a) to (c), $\bar{V}(-\infty, t_N)$ exists for any nonstochastic, temporal variation in the model and sampling.

Although $\bar{V}(-\infty, t_N)$ is independent of Q, it is generally dependent on t_N. We now prove that if the model and sampling exhibit periodic variation, then, for each t_N, $\bar{V}(-\infty, t_N)$ is an element in a limiting, periodic cycle, $\{\bar{V}_i\}_{i=1}^{\rho}$. In the focal case, in which mixed-frequency data induce periodic sampling, periodic variation in KF iterations arises in $\Lambda(t)$: $[\Lambda(t_1), \ldots, \Lambda(t_\rho), \Lambda(t_{\rho+1}), \ldots, \Lambda(t_{2\rho}), \ldots] = [\Lambda_1, \ldots, \Lambda_\rho, \Lambda_1, \ldots, \Lambda_\rho, \ldots]$. More generally, of course, periodic variation in KF iterations may also arise in F, G, Δ, and Σ_ζ. In general, for $i = 1, \ldots, \rho$, let T_i be the set of sampling times from $-\infty$ to ∞ in the i-th period of the model-sampling cycle. In the focal case, T_i is the set of $t \in (-\infty, \infty)$ for which $\Lambda(t) = \Lambda_i$. Then, $T = \cup_{i=1}^{\rho} T_i$ is the set of all sampling times from $-\infty$ to ∞.

Such a partition exists for any periodic variation in the model and sampling. Let ρ_m = periodicity in model, ρ_s = periodicity in sampling, $\rho_1 = \min[\rho_m, \rho_s]$, $\rho_2 = \max[\rho_m, \rho_s]$, and ρ = overall periodicity used to construct T. Then, $\rho = \rho_2$ if $\rho_2/\rho_1 = \text{int}[\rho_2/\rho_1]$, where int[•] is the integral part of a number, and $\rho = \rho_m \cdot \rho_s$ otherwise.

We now consider KF iterations over sampling times in T_i. Because consecutive sampling times in T_i are ρ periods apart, we need to modify (3.7) for this purpose. First, after eliminating M(t), K(t), and $\Phi(t)$ using (3.1), (3.4), and (3.6), we can write (3.7) as the Riccati equation, $V(t+1) = GG^T + FV(t)F^T - FV(t)\Delta^T\Lambda(t)^T \cdot \{\Lambda(t)[\Sigma_\zeta + \Delta V(t)\Delta^T]\Lambda(t)^T\}^{-1} \cdot \Lambda(t)\Delta V(t)F^T$. Then, we iterate $\rho - 1$ times on the Riccati equation, starting with any $t \in T_i$, to obtain the time-invariant mapping $V(t+\rho) = \Pi_{i,\rho}[V(t)]$. $\Pi_{i,\rho}$ is time invariant because it embodies

any periodic variation in the model and sampling. Despite its time invariance, $\Pi_{i,\rho}$ generally varies with i.

To see these properties of $\Pi_{i,\rho}$, consider the case of the application, as discussed at the end of section 2. In this case, $T = \{\ldots, -1, 0, 1, \ldots\} = \cup_{i=1}^{3} T_i$, where T_i is the set of i-th months of quarters. Because t = 1 is the first month of a quarter, $T_1 = \{\ldots, -5, -2, 1, 4, 7, \ldots\}$, $T_2 = \{\ldots, -4, -1, 2, 5, 8, \ldots\}$, and $T_3 = \{\ldots, -3, 0, 3, 6, 9, \ldots\}$. Because $\Pi_{1,3}$ is formed by starting with $t \in T_1$, $\Pi_{1,3}$ is defined by the sequence $[\Lambda(t), \Lambda(t+1), \Lambda(t+2)] = [\Lambda_1, \Lambda_1, \Lambda_2]$, where $\Lambda_1 = [1, 0]$ and $\Lambda_2 = I_2$. Correspondingly, $\Pi_{2,3}$ and $\Pi_{3,3}$ are defined by the sequences $[\Lambda_1, \Lambda_2, \Lambda_1]$ and $[\Lambda_2, \Lambda_1, \Lambda_1]$.

Consider again sequences $\{V(t_j)\}_{j=1}^{N}$ generated by the Riccati equation, where $\{t_j\}_{j=1}^{N}$ is a sequence of consecutive sampling times in T, t_1 is a variable element of T_i which $\to -\infty$, t_N is a fixed of element T_i, and $V(t_1) = Q \geq 0$. Let $\{t_{ij}\}_{j=1}^{N}$ denote the subsequence of sampling times in $\{t_j\}_{j=1}^{N}$ which are consecutive elements of T_i. Because $\{V(t_{ij})\}_{j=1}^{N}$ is a subsequence of $\{V(t_j)\}_{j=1}^{N}$, $\lim_{t_1 \to -\infty} \bar{V}(t_1, t_N) = \bar{V}(-\infty, t_N)$ implies that $\lim_{t_{i1} \to -\infty} \bar{V}(t_{i1}, t_{iN}) = \bar{V}(-\infty, t_N)$. Because $\{V(t_{ij})\}_{j=1}^{N}$ is generated by the time-invariant mapping $\Pi_{i,\rho}$, the convergence of $\bar{V}(t_{i1}, t_{iN})$ as $t_{i1} \to -\infty$ only depends on $t_{iN} - t_{i1}$; in particular, it does not depend on the locations of t_{i1} and t_{iN} in calender time. Therefore, for any $i = 1, \ldots, \rho$ and fixed $t_{i1} \in T_i$, $\lim_{t_{iN} \to \infty} \bar{V}(t_{i1}, t_{iN}) = \bar{V}(-\infty, t_{iN})$. Evidently, $\bar{V}(-\infty, t_{iN})$ only depends on i. To emphasize this, we write $\bar{V}(-\infty, t_{iN})$ as $\bar{V}_i$. Now, to say that the subsequence $\{V(t_{ij})\}_{j=1}^{\infty}$ converges to $\bar{V}_i$ is equivalent to saying that the full sequence $\{V(t_j)\}_{j=1}^{\infty}$ converges to the periodic cycle $\{\bar{V}_i\}_{i=1}^{\rho}$.

This concludes the proof that if the model and sampling exhibit

periodic variation, with overall periodicity ρ, and assumptions (a) to (c) hold, then, $\{M(t), K(t), \Phi(t), V(t)\}_{t=1}^{\infty}$, generated by KF iterations (3.1), (3.4), (3.6), and (3.7), converges to a periodic cycle, $\{\bar{M}_i, \bar{K}_i, \bar{\Phi}_i, \bar{V}_i\}_{i=1}^{\rho}$. Todd (1989) gave a related proof in an analysis of seasonal economic models formulated as linear optimal regulators.

We now consider asymptotic stability of the KF. Using (2.3), (2.9), (3.2), (3.5), and (3.6), we derive $\tilde{x}(t+1) = \Phi(t)\tilde{x}(t) + Ge(t+1) - K(t)v(t)$. The KF is said to be asymptotically stable if, abstracting from disturbances and observation errors, e and v, the state-prediction error, $\tilde{x}$, approaches 0 as $t \to \infty$. The KF is known to be asymptotically stable if assumptions (a) to (c) hold. This is true for any nonstochastic, temporal variation in the model and sampling. See, e.g., Kwakernaak and Sivan (1972, pp. 534-536).

For the case in which the model and sampling vary periodically, with periodicity ρ, we now characterize asymptotic stability in terms of limiting closed-loop matrices. Analogous to the construction of $\Pi_{i,\rho}$, for $i = 1, \ldots, \rho$ and $t \in T_i$, we iterate $\rho - 1$ times on $\tilde{x}(t+1) = \Phi(t)\tilde{x}(t)$, to obtain $\tilde{x}(t+\rho) = \Theta_{i,\rho}(t)\tilde{x}(t)$, where $\Theta_{i,\rho}(t) = \Phi(t+\rho) \cdots \Phi(t)$. If the KF converges, $\lim_{t\to\infty} \Theta_{i,\rho}(t) = \bar{\Theta}_{i,\rho} = \bar{\Phi}_{i-1} \cdots \bar{\Phi}_1\bar{\Phi}_\rho \cdots \bar{\Phi}_i$. Moreover, in the limit as $t \to \infty$ in T_i, $\tilde{x}(t+\rho k) = (\Theta_{i,\rho})^k\tilde{x}(t)$, for $k = 1, 2, \ldots$. Therefore, when the KF is asymptotically stable, $\bar{\Theta}_{i,\rho}$ is a stable matrix, for $i = 1, \ldots, \rho$. Except in special cases, the eigenvalues of the $\bar{\Theta}_{i,\rho}$'s are difficult to characterize in terms of the eigenvalues of the $\bar{\Phi}_i$'s. For example, when the $\bar{\Phi}_i$'s are all upper- or lower-triangular, the j-th eigenvalue of $\bar{\Theta}_{i,\rho}$ is the product of the j-th eigenvalues of the $\bar{\Phi}_i$'s.

When the model is explosively nonstationary (F has some

eigenvalues on or outside the unit circle) and observations are periodically missing, then, some $\bar{\Phi}_i$'s may be unstable. In such a case, if the KF is asymptotically stable, then, remaining $\bar{\Phi}_i$'s must be sufficiently stable to ensure stable $\bar{\theta}_{i,\rho}$'s. Equation (35) in Verhaegen and Van Dooren (1986) shows that numerical stability of the KF will be more fragile in such cases because, within the sampling cycle, unstable $\bar{\Phi}_i$'s may cause computational errors to build up to dangerous or catastrophic levels before they can be mitigated by stable $\bar{\Phi}_i$'s.

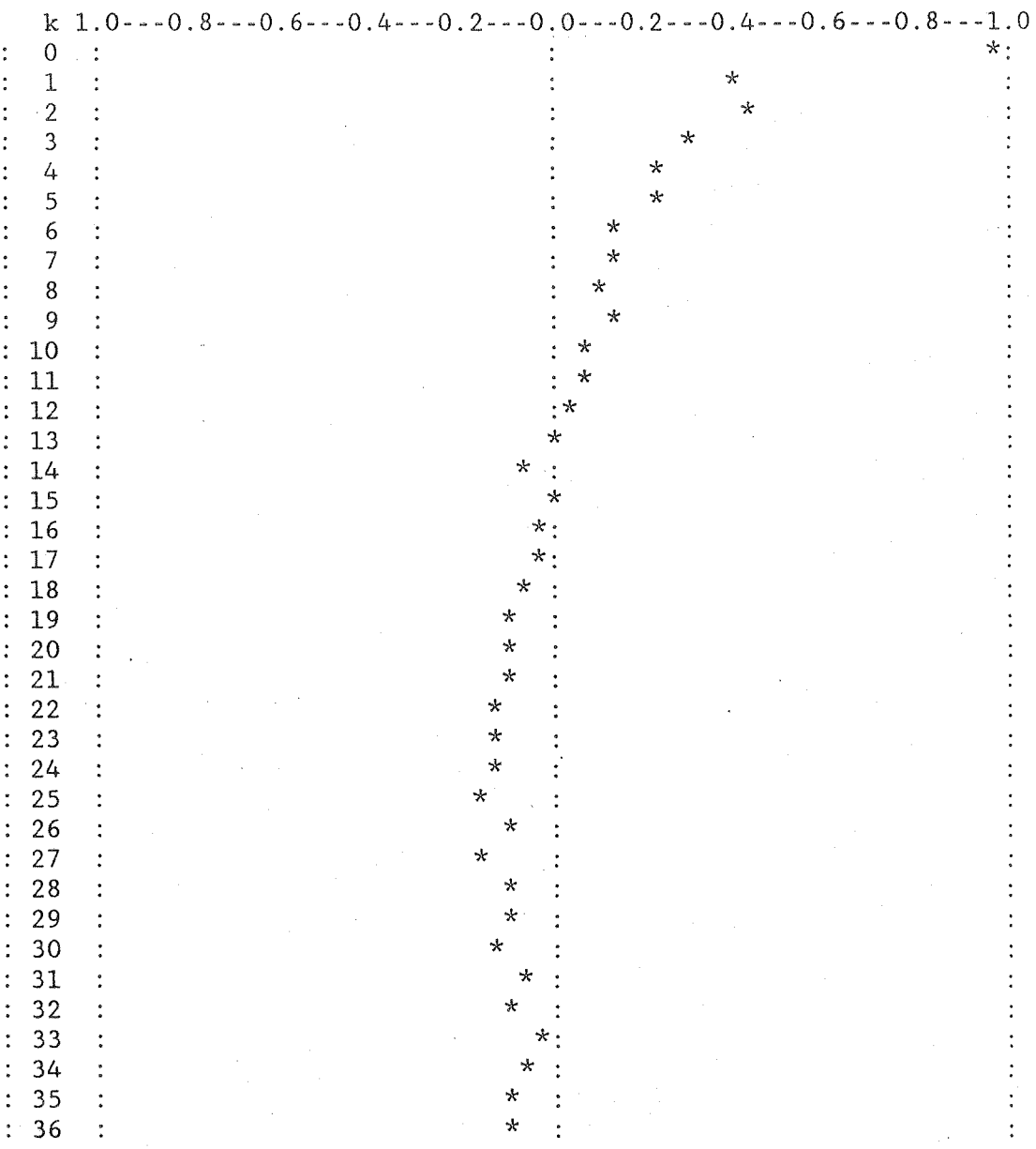
To conclude, we consider restrictions on parameters under which x is controllable and effectively reconstructible when the model and sampling are as in the application. We only consider some sufficient conditions and do not attempt to investigate necessary conditions. To see how the sampling scheme complicates the restrictions, we first state restrictions for the case of a bivariate ARMA(1,1) model whose variables are stocks observed every period. In this case, R(2) has full rank (4), hence, x is reconstructible, for all parameter values, and, C(2) has full rank, hence, x is controllable, if (i) $|AB + C| \neq 0$ and (ii) $|C| \neq 0$, where, for simplicity, A_1, B_0, and B_1 are denoted as A, B, and C.

Generally, we want reconstructibility for the convergence proof because it ensures that $\bar{V}(t_1,t_N)$ is bounded from above as $t_1 \to -\infty$. When the data are sampled as in the application, columns 3 and 6 of R(t) are identically equal to zero for all t, so that state variables 3 and 6, x_3 and x_6, are unreconstructible. However, elements of $\bar{V}(t_1,t_N)$ associated with x_3 and x_6 are bounded as long as the remaining state variables, x_1, x_2, x_4, and x_5, are reconstructible. This follows for two reasons. First, when x_1, x_2, x_4, and x_5 are reconstructible, elements of $\bar{V}(t_1,t_N)$ associated with these state variables are bounded. Second, as an

analysis of the state representation reveals, x_3 and x_6, are just cumulators of lagged values of flows, x_2 and x_5, and have no active role to play in the update of V(t) with the Riccati equation. Columns 1, 2, 4, and 5 of R(3) have full rank, so that x_1, x_2, x_4, and x_5 are reconstructible, if (iii) $A_{12} \neq 0$, where A_{12} is the (1,2) element of A.

In the situation in the application, C(3) has full rank (6), hence, x is controllable, if $|C| \neq 0$, (iv) $B_{21} \neq 0$, (v) $B_{22} \neq 0$, (vi) $(A_{21} - A_{11})B_{11}B_{22} + B_{22}(C_{21} - C_{11}) + B_{21}(C_{12} - C_{22}) \neq 0$, and (vii) $A_{11}B_{11}B_{22}(A_{11}B_{21} - A_{21}B_{11}) + A_{11}B_{21}(B_{11}C_{22} - B_{21}C_{12} + B_{22}C_{11}) - B_{11}B_{22}(A_{11}C_{21} + A_{21}C_{11}) + C_{11}(B_{21}C_{22} - B_{22}C_{21}) \neq 0$. Evidently, restrictions (ii) to (vii) are satisfied by the best model reported in Table 1.

Figure 1. Monthly Serial Correlations of Employment.

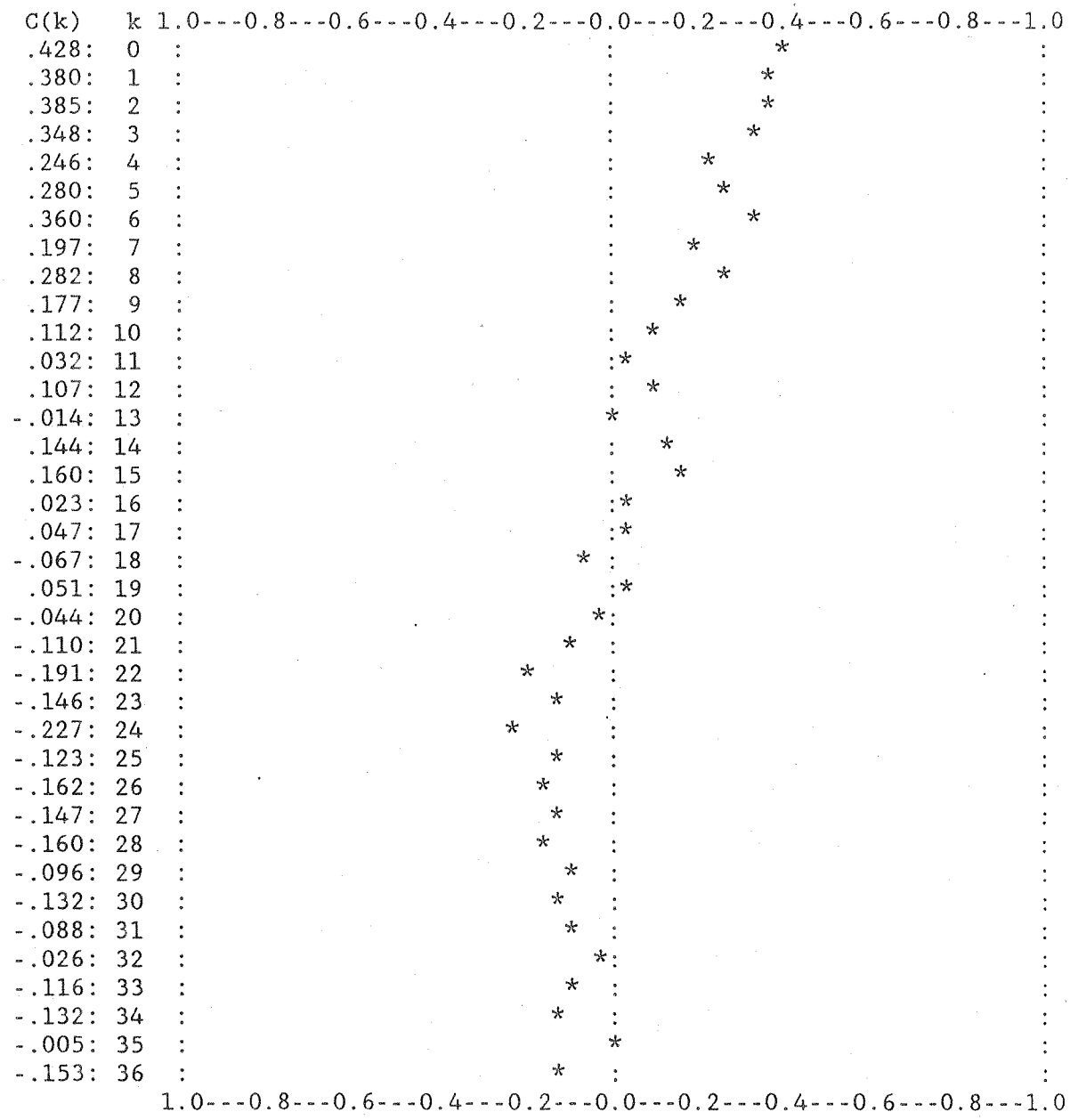


a. Figures 2 to 9 are based on $y_1(t)$ = mean-adjusted, annualized, monthly, first-difference of the natural logarithm of employment, for January 1958 to December 1978, and $y_2(t)$ = mean-adjusted, annualized, quarterly, first-difference of the natural logarithm of real GNP, for quarter 1, 1958 to quarter 4, 1988.

b. $C(k) = Cor[y_1(t), y_1(t-k)]$. Asymptotic standard errors of C(k)'s range from .063 to .068. In figures 1 to 8, asymptotic standard error of $C(k) = N(k)^{-.5}$, where N(k) = number of sampling times used to compute C(k).

c. Ljung-Box Q statistic = 224. Marginal significance level of Q = $.538 \times 10^{-28}$, at 36 degrees of freedom.

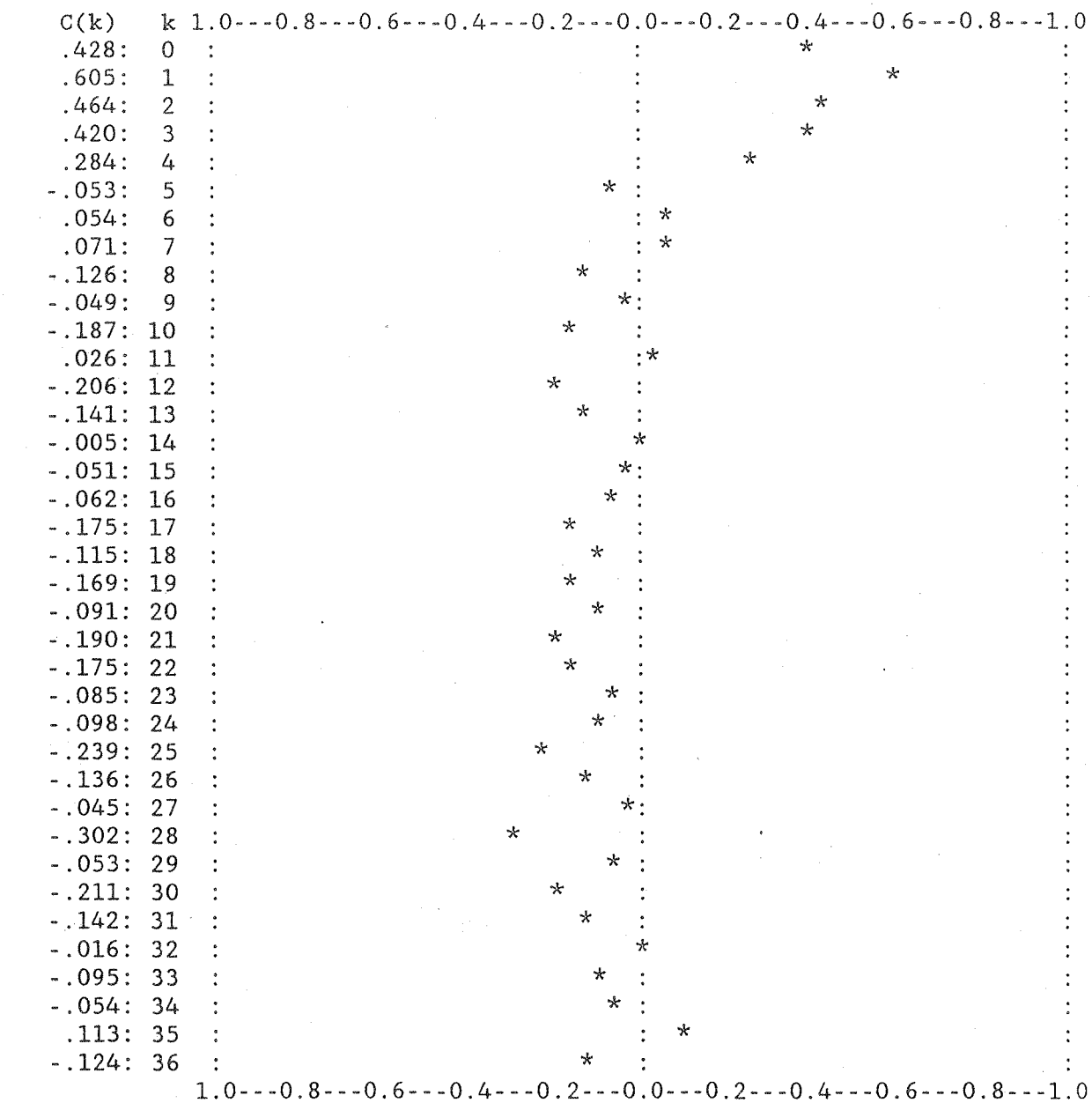
Figure 2. Monthly Cross-Serial Correlations of Employment and GNP.



a. C(k) = Cor[$y_1(t)$, $y_2(t-k)$]. Asymptotic standard errors of C(k)'s range from .109 to .118.

b. Ljung-Box Q statistic = 99.6. Marginal significance level of Q = .718 x 10^{-7}, at 36 degrees of freedom.

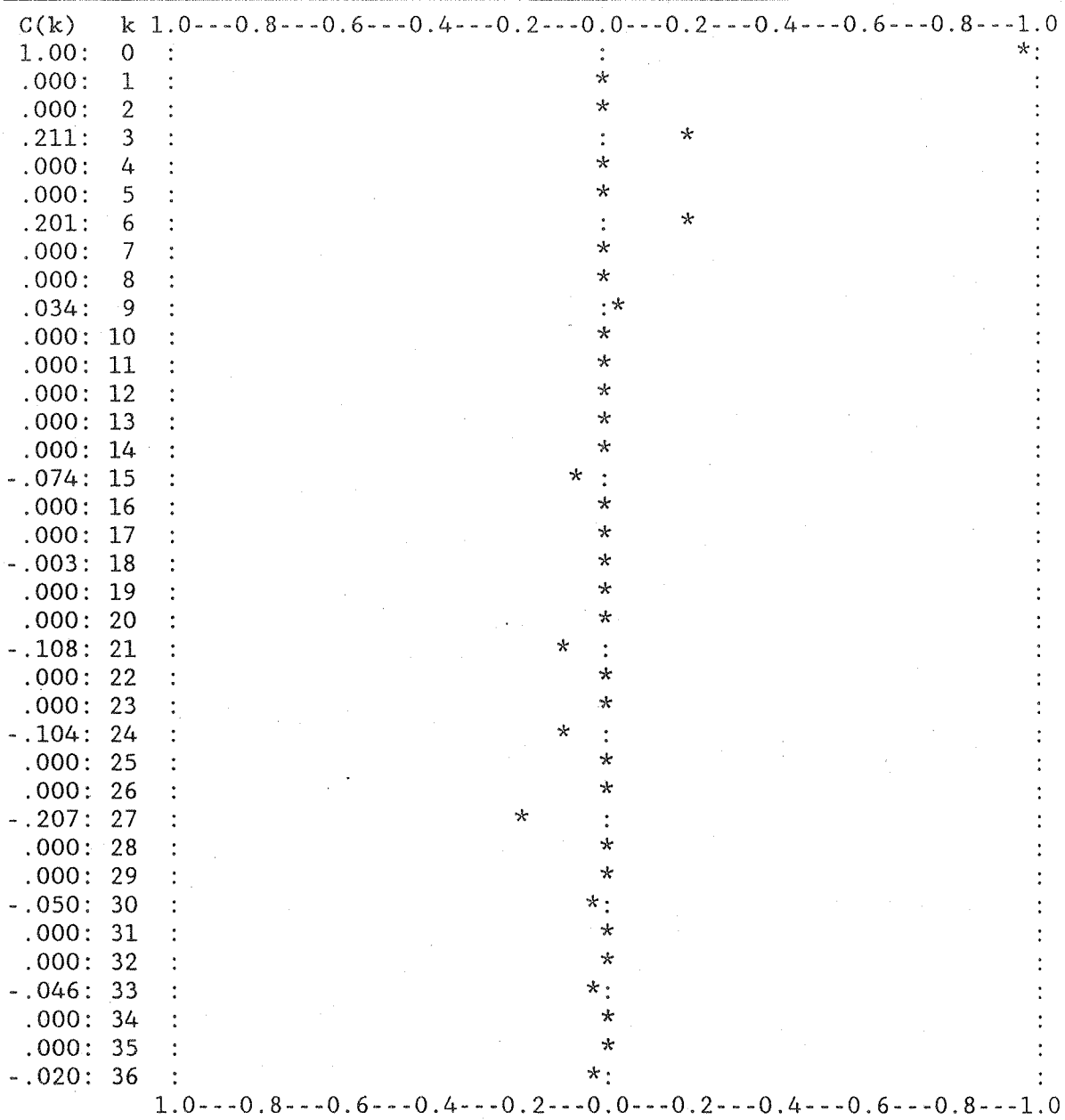
Figure 3. Monthly Cross-Serial Correlations of GNP and Employment.



a. C(k) = Cor[$y_2(t)$, $y_1(t-k)$]. Asymptotic standard errors of C(k)'s range from .109 to .118.

b. Ljung-Box Q statistic = 115. Marginal significance level of Q = .248 x 10^{-9}, at 36 degrees of freedom.

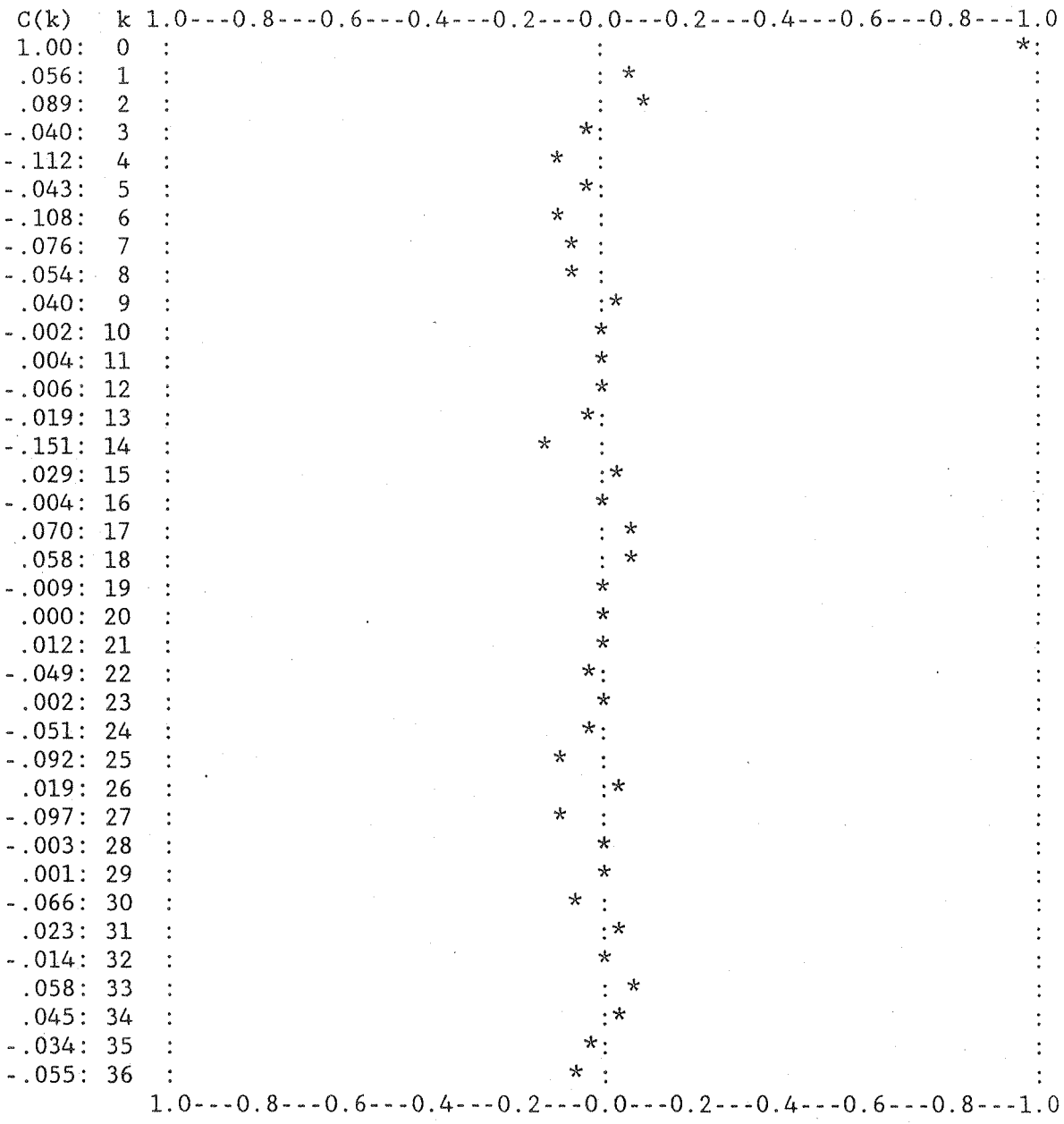
Figure 4. Monthly Serial Correlations of GNP.



a. C(k) = E[y_2(t), y_2(t-k)], where, in figures 4 and 8, C(k) = 0 when first three digits are insignificant or k is at non-quarterly lag at which C(k) cannot be computed. Asymptotic standard errors of C(k)'s range from .110 to .118.

b. Ljung-Box Q statistic = 12.8. Marginal significance level of Q = .381, at 12 degrees of freedom (only 12 quarterly C(k)'s can be computed within 36 monthly lags).

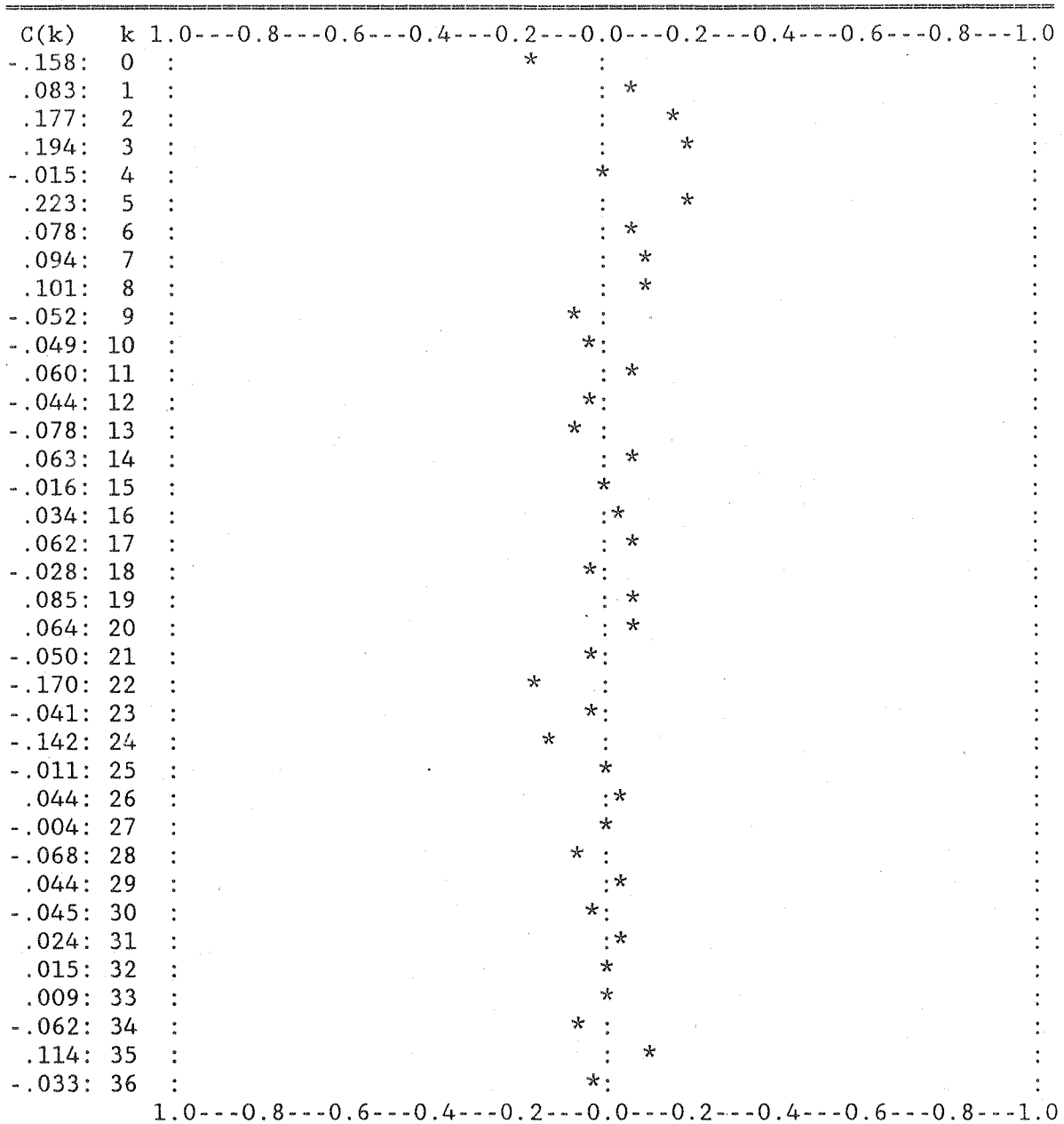
Figure 5. Monthly Serial Correlations of Innovations of Employment.



a. C(k) = Cor[$\xi_1(t)$, $\xi_1(t-k)$], where, in figures 5 to 8, $\xi_1(t)$ = innovation of $y_1(t)$ and $\xi_2(t)$ = innovation of $y_2(t)$. Asymptotic standard errors of C(k)'s range from .063 to .068.

b. Ljung-Box Q statistic = 28.4. Marginal significance level of Q = .603, at 31 degrees of freedom. In figures 5 to 7, Q is evaluated at 31 = 36 - 5 degrees of freedom, where 5 = integral part of 11 ÷ 2 is meant to represent "average" loss of degrees of freedom in an innovation series due to estimation of 11 parameters.

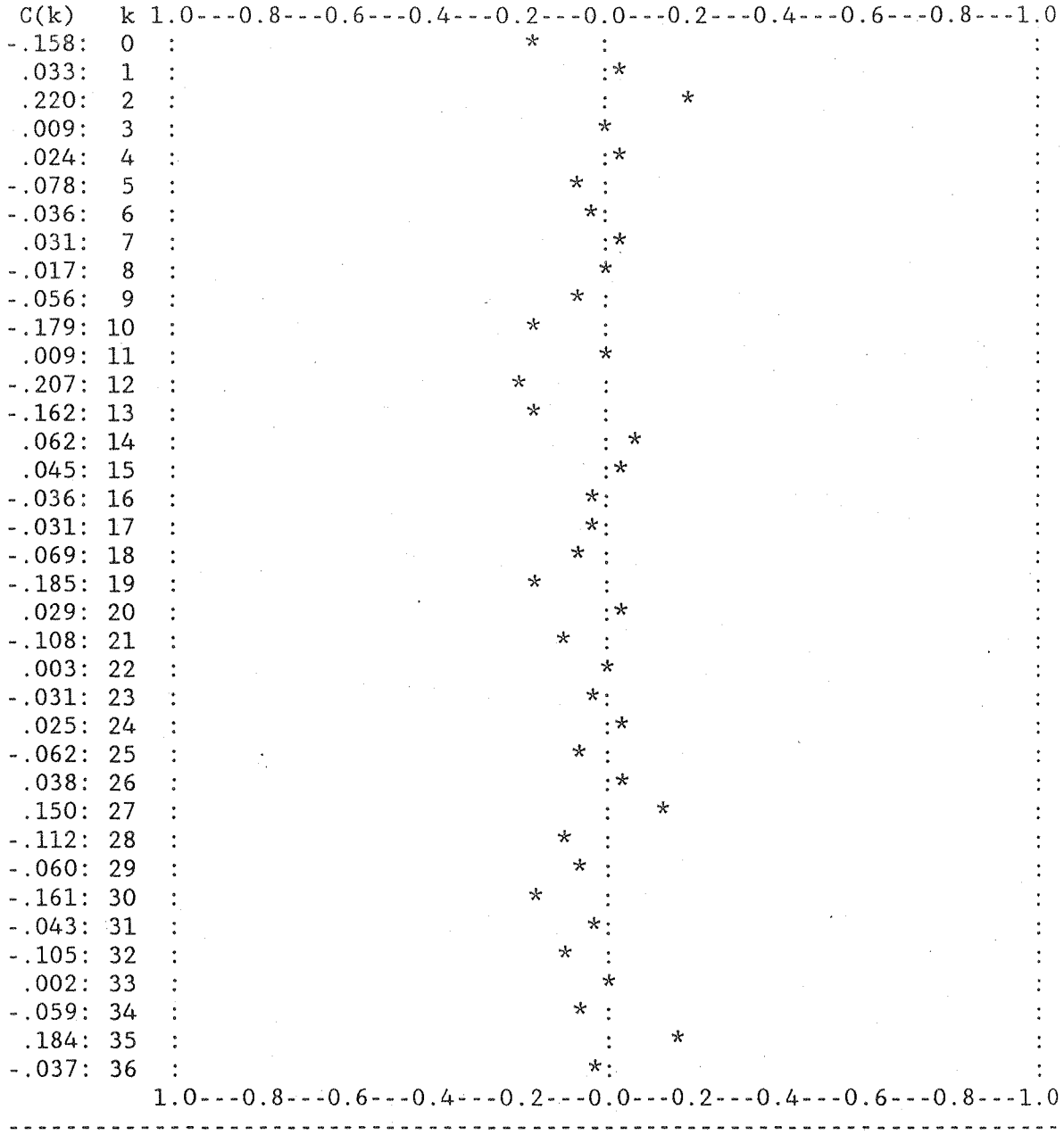
Figure 6. Monthly Cross-Serial Correlations of Innovations of Employment and GNP.



a. $C(k) = Cor[\xi_1(t), \xi_2(t-k)]$. Asymptotic standard errors of C(k)'s range from .109 to .118.

b. Ljung-Box Q statistic = 21.8. Marginal significance level of Q = .890, at 31 degrees of freedom.

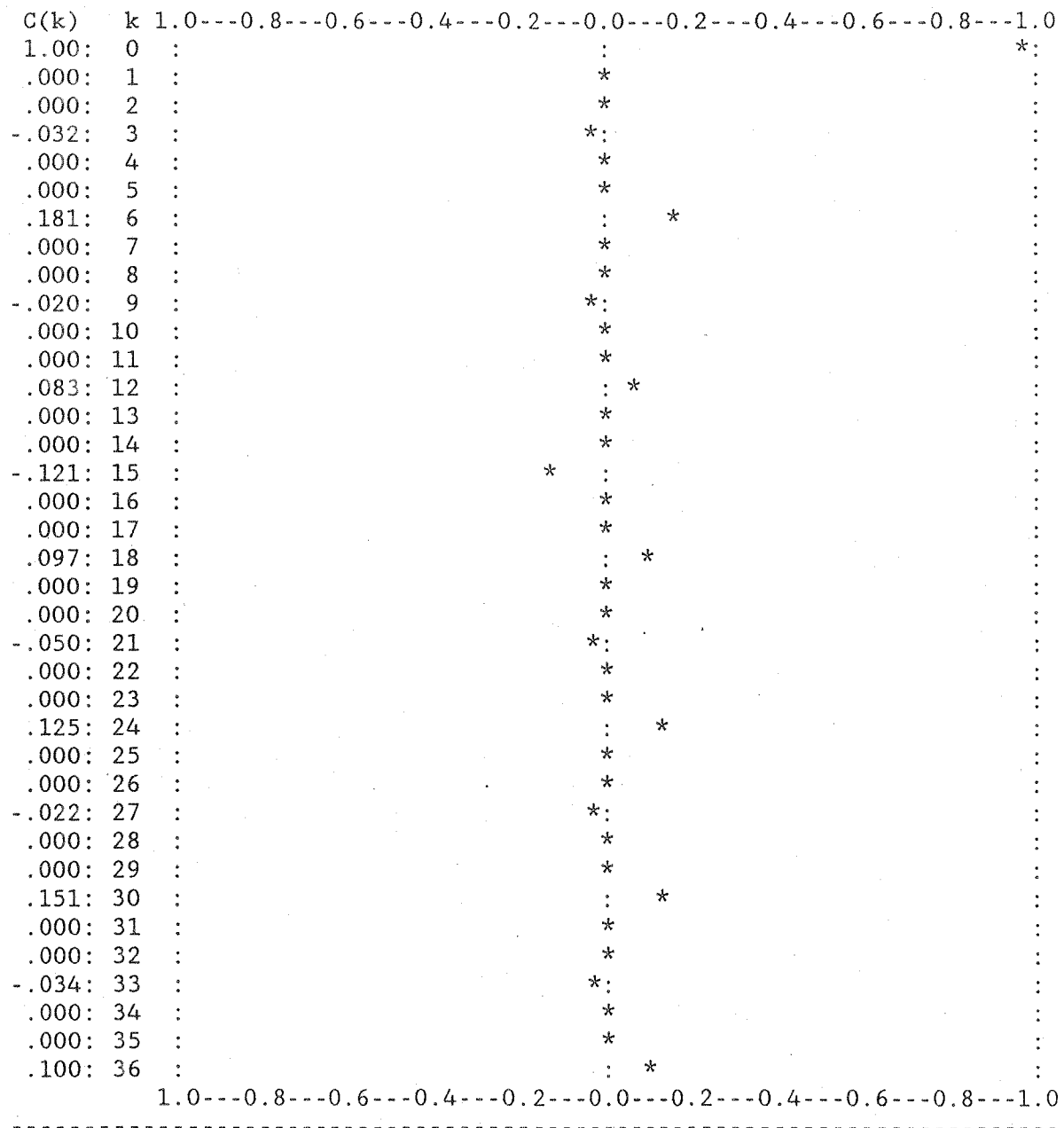
Figure 7. Monthly Cross-Serial Correlations of Innovations of GNP and Employment.



a. $C(k) = Cor[\xi_2(t), \xi_1(t-k)]$. Asymptotic standard errors of C(k)'s range from .109 to .118.

b. Ljung-Box Q statistic = 27.1. Marginal significance level of Q = .669, at 31 degrees of freedom.

Figure 8. Monthly Serial Correlations of Innovations of GNP.



a. C(k) = Cor[$\xi_2(t)$, $\xi_2(t-k)$]. Asymptotic standard errors of C(k)'s range from .110 to .118.

b. Ljung-Box Q statistic = 9.89. Marginal significance level of Q = .195. Q is evaluated at 7 = 12 - 5 degrees of freedom, where, as in figures 5 to 7, 5 = "average" loss of degrees of freedom in the innovation series (only 12 quarterly C(k)'s can be computed within 36 monthly lags).

Table 1. Best Overall Bivariate Employment-GNP Model: Unrestricted ARMA(1,1).

1. Maximum likelihood estimates:

parameter	estimate	asy. std. err.	t-ratio
1	.799	.455	1.75
2	.203	.170	1.19
3	.417	1.11	.377
4	.353	.0003	1228.
5	2.37	.934	2.54
6	.634	4.52	.140
7	1.34	6.01	.222
8	-.615	3.01	-.204
9	1.72	3.89	.441
10	-.697	6.20	-.112
11	-.613	7.89	-.078

2. Implied ARMA coefficient matrices:

$$A_1 = \begin{bmatrix} .799 & .417 \\ .203 & .353 \end{bmatrix}, \quad B_0 = \begin{bmatrix} 2.37 & 0.00 \\ .634 & 1.34 \end{bmatrix},$$

$$B_1 = \begin{bmatrix} -.615 & -.697 \\ 1.72 & -.613 \end{bmatrix}.$$

3. AR and MA characteristic roots:

AR roots = .942, .209,
MA roots = .289 ± i × .643 (modulus = .705).

4. Model-fit summary statistics:

variable	std. err.	R^2
employment	2.83	.228
GNP	2.82	.472

5. Bias-corrected AIC = 1231.

Estimation period: January 1958 to December 1978.

Table 2. Forecast-Evaluation Statistics of Model in Table 1.

1. Employment forecast-evaluation statistics:

months ahead	std. err.	Theil U
1	2.33	.576
2	2.40	.592
3	2.56	.632
4	2.69	.664
5	2.77	.684
6	2.83	.699
7	2.87	.710
8	2.93	.724
9	3.02	.747
10	3.10	.767
11	3.18	.786
12	3.22	.796

2. GNP forecast-evaluation statistics:

months ahead	std. err.	Theil U
1	2.51	.437
2	2.79	.485
3	3.34	.582
4	3.72	.647
5	3.85	.670
6	3.97	.692
7	4.20	.731
8	4.14	.721
9	4.15	.723
10	4.24	.738
11	4.25	.740
12	4.30	.749

a. Forecast-evaluation period: January 1979 to December 1988.
b. Standard errors of forecasts are synonymous with root-mean-squared forecast errors.
c. Theil U statistic = standard error of forecast ÷ standard error of the naive forecast, where the naive forecast is the most recent observation at the time the forecast is made.

Table 3. Comparison with Other Forecasts for Comparable Periods.

1. Quarterly GNP forecast standard errors:

quarters ahead	Table 2	Roberds	McNees
1	3.34	3.87	4.25
2	3.97	3.85	3.61
3	4.15	4.15	3.78
4	4.30	4.34	3.82

2. Monthly GNP forecast standard errors:

months ahead	Table 2	Trehan
1	2.51	1.81
2	2.79	3.34

a. Column marked "Roberds" is from Table 2, part 1, column 2 in Roberds (1988) and reflects forecasts over quarter 1, 1977 to quarter 1, 1987.
b. Column marked "McNees" is the average of columns 1 to 4 in Table 2 in McNees (1986) and reflects forecasts over quarter 2, 1980 to quarter 1, 1985.
c. Column marked "Trehan" is from Table 2, part 2, column 3 in Trehan (1989) and reflects forecasts over quarter 4, 1978 to quarter 4, 1988.

Table 4. Best Univariate Employment Model: AR(1).

1. Maximum likelihood estimates:

parameter	estimate	asy. std. err.	t-ratio
A_1	.955	.0876	10.9
B_0	2.34	.548	4.27

2. Model-fit summary statistics:

standard error of innovation = 2.82,
R^2 = .235.

3. Bias-corrected AIC = 785.

4. Serial correlations of innovations:

Ljung-Box Q = 25.3:
degrees of freedom = 34,
marginal significance = .860.

5. Forecast-evaluation statistics:

months ahead	std. err.	Theil U
1	2.36	.582
2	2.41	.596
3	2.58	.637
4	2.70	.667
5	2.79	.689
6	2.85	.705
7	2.91	.718
8	2.96	.731
9	3.05	.754
10	3.13	.773
11	3.20	.791
12	3.24	.801

a. Estimation period: January 1958 to December 1978.
b. Forecast-evaluation period: January 1979 to December 1988.

Table 5. Best Univariate GNP Model: AR(1).

1. Maximum likelihood estimates:

parameter	estimate	asy. std. err.	t-ratio
A_1	.936	.252	3.71
B_0	3.89	1.89	2.06

2. Model-fit summary statistics:

   standard error of innovations = 3.97,
   R^2 = -.079.

3. Bias-corrected AIC = 409.

4. Serial correlations of innovations:

   Ljung-Box Q = 5.23:
   degrees of freedom = 10,
   marginal significance = .876.

5. Forecast-evaluation statistics:

quarters ahead	std. err.	Theil U
1	3.81	.708
2	4.07	.757
3	4.29	.796
4	4.31	.800

a. Estimation period: quarter 1, 1958 to quarter 4, 1978.
b. Forecast-evaluation period: quarter 1, 1979 to quarter 4, 1988.

Table 6. Best Standard-Method Employment-GNP Model: Restricted AR(1).

1. Maximum likelihood estimates:

parameter	estimate	asy. std. err.	t-ratio
1	.938	.112	8.40
2	.387	.242	1.60
3	2.31	.663	3.49
4	1.11	2.47	.452
5	1.78	1.03	1.72

2. Implied ARMA coefficient matrices:

$$A_1 = \begin{bmatrix} .938 & 0.00 \\ .387 & 0.00 \end{bmatrix}, \quad B_0 = \begin{bmatrix} 2.31 & 0.00 \\ 1.11 & 1.78 \end{bmatrix},$$

3. Model-fit summary statistics:

variable	std. err.	R^2
employment	2.84	.227
GNP	3.24	.304

4. Bias-corrected AIC = 1246.

5. Own- and cross-serial correlations of innovations:

variables	Ljung-Box Q	deg. free.	marg. sign.
own-ser: emp	24.6	34	.881
crs-ser: emp-gnp	23.5	34	.911
crs-ser: gnp-emp	47.0	34	.0689
own-ser: gnp	7.74	10	.655

a. Estimation period: January 1958 to December 1978.
b. Zero-valued coefficients in A_1 and B_0 were imposed during estimation.

Table 7. Forecast-Evaluation Statistics of Model in Table 6.

1. Employment forecast-evaluation statistics:

months ahead	std. err.	Theil U
1	2.36	.583
2	2.42	.598
3	2.58	.637
4	2.70	.667
5	2.79	.689
6	2.85	.704
7	2.91	.718
8	2.96	.732
9	3.05	.753
10	3.11	.769
11	3.18	.786
12	3.22	.796

2. GNP forecast-evaluation statistics:

months ahead	std. err.	Theil U
1	2.96	.516
2	3.17	.552
3	3.51	.612
4	3.79	.660
5	3.91	.680
6	4.01	.698
7	4.24	.738
8	4.18	.727
9	4.18	.728
10	4.27	.744
11	4.27	.744
12	4.32	.752

a. Forecast-evaluation period: January 1979 to December 1988.
b. Standard errors of forecasts are synonymous with root-mean-squared forecast errors.
c. Theil U statistic = standard error of forecast ÷ standard error of the naive forecast, where the naive forecast is the most recent observation at the time the forecast is made.

REFERENCES

Akaike, H. (1973), "Information Theory and an Extension of the Maximum Likelihood Principle," pp. 267-281 in Second International Symposium on Information Theory, B. N. Petrov and F. Csaki (eds.), Budapest: Akademia Kiado.

Anderson, B. D. O. and J. B. Moore (1979), Optimal Filtering, Englewood Cliffs, NJ: Prentice Hall.

Ansley, C. F. and R. Kohn (1983), "Exact Likelihood of Vector Autoregressive Moving-Average Process with Missing or Aggregated Data," Biometrika 70: 275-278.

Ansley, C. F. and R. Kohn (1985), "Estimation, Filtering, and Smoothing in State Space Models with Incompletely Specified Initial Conditions," The Annals of Statistics 13: 1286-1316.

Bell, W. R. and S. Hillmer (1987), "Initiallizing the Kalman Filter in the Nonstationary Case: with Applications to Signal Extracton," Research Report, Statistical Research Division, U.S. Bureau of the Census, Washington, DC.

Bordignon, S. and U. Trivellato (1989), "The Optimal Use of Provisional Data in Forecasting with Dynamic Models," Journal of Business and Economic Statistics 7: 275-286.

Burmeister, E., K. D. Wall, and J. D. Hamilton (1986), "Estimation of Unobserved Expected Monthly Inflation Using Kalman Filtering," Journal of Business and Economic Statistics 4: 147-160.

Christiano, L.J. (1985), "A Method for Estimating the Timing Interval in a Linear Econometric Model, with an Application to Taylor's Model of Staggered Contracts," Journal of Economic Dynamics and Control 9: 363-404.

Carson, C. S. (1987), "GNP: An Overview of Source Data and Estimating Methods," Survey of Current Business 67 (July): 103-126.

Corrado, C. and M. Greene (1988), "Reducing Uncertainty in Short-Term Projections: Linkage of Monthly and Quarterly Models," Journal of Forecasting 7: 77-102.

Conrad, W. and C. Corrado (1979), "Application of the Kalman Filter to Revisions in Monthly Retail Sales Estimates," Journal of Economic Dynamics and Control 1: 177-198.

Doan, T. A., R. B. Litterman, and C. A. Sims (1984), "Forecasting and Conditional Projections Using Realistic Prior Distributions," Econometric Reviews 3: 1-100.

Dongarra, J. J., J. R. Bunch, C. B. Moler, and G. W. Stewart (1979), LINPACK User's Guide, Philadelphia, PA: SIAM.

Engle, R. F. and C. W. J. Granger (1987), "Cointegration and Error Correction: Representations, Estimation, and Testing," Econometrica 55: 251-276.

Findley, D. F. and C. Z. Wei (1989), "Beyond Chi-Squared: Likelihood Ratio Procedures for Comparing Non-Nested, Possibly Incorrect Regressors," Working Paper, Statistical Research Division, U.S. Bureau of the Census, Washington, DC.

Fitzgerald, T. J. and P. J. Miller (1989), "A Simple Way to Estimate Current-Quarter GNP," Federal Reserve Bank of Minneapolis Quarterly Review 13 (Fall): 27-31.

Fountis, N. G. and D. A. Dickey (1989), "Testing for a Unit Root Nonstationarity in Multivariate Autoregressive Time Series," The Annals of Statistics 17: 419-428.

Golub, G. H. and C. F. Van Loan (1983), Matrix Computations, Baltimore, MD: Johns Hopkins University Press.

Hannan, E. J. (1979), "The Statistical Theory of Linear Systems," pp. 83-121 in Developments in Statistics, vol. 2, P. R. Krishnaiah (ed.), Orlando, FL: Academic Press.

Hansen, L. P. and T. J. Sargent (1980), "Formulating and Estimating Dynamic Linear Rational Expectations Models," Journal of Economic Dynamics and Control 2: 7-46.

Hansen, L. P. and T. J. Sargent (1983), "The Dimensionality of the Aliasing Problem in Models with Rational Spectral Densities," Econometrica 51: 377-387.

Harvey, A. C. and R. G. Pierse (1984), "Estimating Missing Observations in Economic Time Series," Journal of the American Statistical Association 79: 125-131.

Howrey, E. P. (1984), "Data Revision, Reconstruction, and Prediction: An Application to Inventory Investment," Review of Economics and Statistics 66: 386-393.

Hurvich C. M. and C. L. Tsai (1989), "Regression and Time Series Model Selection in Small Samples," Biometrika 76: 297-307.

Jones, R. H. (1980), "Maximum Likelihood Fitting of ARMA Models to Time Series with Missing Observations," Technometrics 22: 389-395.

Judge, G. G., W. E. Griffiths, R. C. Hill, and T. C. Lee (1980), The Theory and Practice of Econometrics, New York, NY: John Wiley.

Kohn, R. and C. F. Ansley (1982), "A Note on Obtaining the Theoretical Autocovariances of an ARMA Process," Journal of Statistical Computation and Simulation 15: 273-283.

Kwakernaak, H. and R. Sivan (1972), Linear Optimal Control Systems, New York: Wiley-Interscience.

Litterman, R. B. (1983), "A Random Walk, Markov, Model for the Distribution of Time Series," Journal of Business and Economic Statistics 1: 169-173.

Ljung, G. M. and G. E. P. Box (1978), "On a Measure of Fit in Time Series Models," Biometrika 65: 297-303.

Mankiw, N. G. and M. D. Shapiro (1986), "News or Noise: An Analysis of GNP Revisions," Survey of Current Business 66 (May): 20-25.

McNees, S. K. (1986), "Forecasting Accuracy of Alternative Techniques: A Comparison of U.S. Macroeconomic Forecasts," Journal of Business and Economic Statistics 4: 5-15.

More, J. J., B. S. Garbow, and K. E. Hillstrom (1980), "User Guide for MINPACK-1," Report ANL-80-74, Argonne National Laboratory, Argonne, IL.

Morf, M. and T. Kailath (1975), "Square-Root Algorithms for Least-Squares Estimation," IEEE Transactions on Automatic Control AC-20: 487-497.

Mork, K. A. (1987), "Ain't Misbehavin': Forecast Errors and Measurement Errors in Early GNP Estimates," Journal of Business and Economic Statistics 5: 165-175.

Neudecker, H. (1969), "Some Theorems on Matrix Differentiation with Special Reference to Kronecker Matrix Products," Journal of the American Statistical Association 64: 953-963.

Roberds, W. T. (1988), "A Quarterly Bayesian VAR Model of the U.S. Economy," Working Paper, Research Department, Federal Reserve Bank of Atlanta, Atlanta, GA.

Scadding, J. (1987), "The Nature of GNP Revisions," Working Paper, Research Department, Federal Reserve Bank of Cleveland, Cleveland, OH.

Scadding, J. (1988), "Getting the Noise Out: Filtering Early GNP Estimates," Federal Reserve Bank of Cleveland Economic Review 24 (Summer): 24-31.

Schweppe, F. (1965), "Evaluation of Likelihood Function for Gaussian Signals," IEEE Transactions on Information Theory 11: 61-70.

Smith, B. T., J. M. Boyle, J. J. Dongarra, B. S. Garbow, Y. Ikebe, V. C. Klema, and C. B. Moler (1976), Matrix Eigensystem Routines - EISPACK Guide, Lecture Notes in Computer Science, Vol. 6, New York, NY: Springer-Verlag.

Todd, R. M. (1989), "Periodic Linear-Quadratic Methods for Modelling Seasonality", Staff Report No. 127, Research Department, Federal Reserve Bank of Minneapolis, Minneapolis, MN.

Trehan, B. (1989) "Forecasting Growth in Current Quarter Real GNP," Federal Reserve Bank of San Francisco Economic Review (Winter): 39-51.

Verhaegen, M. and P. Van Dooren (1986), "Numerical Aspects of Different Kalman Filter Implementations," IEEE Transactions on Automatic Control AC-31: 907-917.

Whiteman, C. H. and W. T. Roberds (1989), "Monetary Aggregates as Monetary Targets: A Statistical Investigation," Working Paper, Research Department, Federal Reserve Bank of Atlanta, Atlanta, GA.

Young, A. H. (1987), "Evaluation of the GNP Estimates," Survey of Current Business 67 (August): 18-42.

Zadrozny, P. A. (1988), "Gaussian Likelihood of Continuous-Time ARMAX Models when Data are Stocks and Flows at Different Frequencies," Econometric Theory 4: 108-124.

Zadrozny, P. A. (1989), "Analytic Derivatives for Estimation of Linear Dynamic Models," Computers and Mathematics with Applications 18: 539-553.